



A comprehensive review on rare earth elements: resources, technologies, applications, and prospects

Tian-Yu Zhao, Wei-Lun Li*, Sadan Kelebek*, Yeonuk Choi, Cheng-Qian Wu,
Wen-Juan Zhang, Chen-Yang Wang, Zhong-Wei Zhao, Farzaneh Sadri

Received: 27 December 2024 / Revised: 21 May 2025 / Accepted: 21 May 2025 / Published online: 3 August 2025
© Youke Publishing Co., Ltd. 2025

Abstract Rare earth elements (REEs), with their unique magnetic, optical, and electrical properties, have become indispensable strategic resources. **Widely applied in critical fields such as aviation, telecommunications, electronics, energy, transportation, and medicine, REEs play a vital role in advancing technology and driving social and economic development.** However, the REE industry faces numerous challenges, including unbalanced resource distribution, supply and demand imbalances, international competition, technological limitations, and associated environmental pollution. This paper, incorporating both the historical evolution and current state of the REE industry, provides a comprehensive examination of the chemistry, applications, resources, technologies, challenges, and prospects of REEs. Specifically, it analyzes China's REE industry, which holds the largest global reserves and production capacity. As a key feature, this paper introduces the Tai Chi model for sustainable development in the REE industry, offering an in-depth analysis of two primary

approaches—mining and recycling; the four critical participants—governments, enterprises, researchers, and consumers; and the eight essential influencing factors—resources, energy, environment, policy, applications, technology, supply and demand, and economy. The Tai Chi model not only clarifies the responsibilities and significance of each individual but also highlights their interconnectedness, providing a compelling framework for envisioning the sustainable development of the REE industry. Moreover, the paper identifies the major challenges currently facing the industry and offers insights into the future development of REEs. As such, this work contributes to a deeper understanding of the multifaceted REE landscape and underscores the importance of sustainable practices to ensure REEs' lasting positive impact on the global industry.

Keywords Rare earth elements; Resource; Technology; Mining; Recycling; Sustainability

T.-Y. Zhao, S. Kelebek*, Y. Choi, C.-Y. Wang, F. Sadri
The Robert M. Buchan Department of Mining, Queen's
University, Kingston, ON K7L3N6, Canada
e-mail: sadan.kelebek@queensu.ca

W.-L. Li*
Zhongyuan Critical Metals Laboratory, School of Materials
Science and Engineering, Zhengzhou University, Zhengzhou
450001, China
e-mail: liweilun@zzu.edu.cn

C.-Q. Wu
Department of Mechanical and Industrial Engineering,
University of Toronto, Toronto, ON M5S 3G8, Canada

W.-J. Zhang, Z.-W. Zhao
School of Metallurgy and Environment, Central South
University, Changsha 410083, China

1 Introduction

Rare earth elements (REEs) are referred to as “the seeds of technology” [1] and “industrial gold” [2]. They refer to a group of elements consisting of 17 elements [3]. Due to their similar physical and chemical properties [4], they are often discussed together and collectively referred to as REEs. Given their excellent optical, electrical, and magnetic properties, they are widely utilized in various applications including permanent magnets, automotive catalytic converters, metal alloys, hybrid engines, glass, ceramics, phosphors, television and computer screens, photocatalysis, petroleum refining, computer hard drives, polishing, medical tracers, and fertilizers [4–10]. They



profoundly impact technological development and social transformation, playing an irreplaceable role as a highly important strategic resource. Therefore, many countries, regions, and international organizations position REEs as critical raw materials [11].

The application of REEs can be traced back to the 1950s when they played an important role in the military and nuclear fields [6, 12]. Today, REEs play a crucial role in modern science and technology. With the continuous development of the global economy and the progress of high-tech industries, the demand for REEs will increase rapidly in the coming decades [13]. Especially in recent years, with increasing awareness of environmental protection and sustainable development, REEs are also widely applied in environmental protection and new energy fields, such as wind turbines, energy-saving light bulbs, and solar panels [14–17]. Additionally, REEs can be employed in improving automobile fuel efficiency and reducing industrial exhaust emissions [18]. Therefore, the world has put forward higher requirements for the production of REEs, necessitating the establishment of a long-term stable supply chain to meet the huge demand [19].

The current production and supply and demand situation of REEs faces a wide range of challenges. Although short-term demand can be met by existing REE mines [20], sustainable production and supply are affected by multiple interrelated factors, including geology/mineralogy, ore grade, mineral reserve tonnage, extraction technology, environmental impact, and policy [19, 21, 22]. Even minor fluctuations in any of these factors can lead to disruptions or crises in REE supply. Broadly, these challenges can be classified into technical and non-technical categories. Technical challenges primarily involve the inherent complexity of REE extraction and processing, such as low natural concentrations of REEs, complex mineralogy, and limited efficiency in current separation and purification technologies. Non-technical challenges encompass a broader range of cross-cutting issues, including economic factors (e.g., high production costs, market volatility), political and geopolitical factors (e.g., export restrictions, trade policies, resource nationalism), social concerns (e.g., public opposition to mining operations), and environmental and regulatory constraints (e.g., ecological impact assessments, permitting delays) [11, 23, 24]. These non-technical issues are equally critical in determining the feasibility and stability of REE supply chains.

Furthermore, as applications of REEs continue to expand across high-tech and clean energy sectors, global demand is rising rapidly, resulting in greater uncertainty in the supply landscape. In response, countries are fiercely competing to secure REE resources and develop technological capabilities, intensifying strategic interests and investment in the REE industry.

Overall, REEs are critical non-renewable strategic resources that have a significant impact on the sustainable advancement of science, technology, and modern civilization [25, 26]. However, their application and development are hindered by a combination of international disputes, supply–demand imbalances, economic constraints, political and social challenges, and environmental concerns. Ensuring the sustainability of the REE industry, therefore requires a comprehensive approach that addresses not only technical factors—such as production efficiency and extraction technologies—but also non-technical aspects, including economic viability, geopolitical stability, and regulatory frameworks.

As a comprehensive and critical review, this manuscript presents a thorough and structured analysis of the current status, development trends, and prospects of REEs. It summarizes various aspects of REEs, including properties, applications, minerals, and environmental impacts, discussing the importance of sustainability from a technical perspective. In addition, it analyzes the impact of international dynamics and political policies on REEs, particularly focusing on China, the world's largest holder and consumer of REEs. Finally, this paper proposes the concept of sustainable production of REEs and constructs a Tai Chi philosophical framework, which emphasizes the complementary relationship between two technical routes of REE production: mining and recycling. It also provides both scientific and philosophical insights into sustainable REE production, articulated through eight interconnected dimensions that underscore the importance of individual factors and their systemic interplay.

2 REE basic information

2.1 Chemistry

The International Union of Pure and Applied Chemistry (IUPAC) defines REEs as 17 elements, including the 15 lanthanides, yttrium, and scandium [7, 27, 28]. The history of REEs spans several centuries, with the term “rare earth” coined following their discovery in 1788 by a miner in Ytterby, Sweden [29, 30]. Yttrium was the first REE to be discovered in 1794 and separated in 1828. Although cerium was discovered after yttrium, it was the first REE to be separated [19, 31, 32]. The discovery and separation of the 17 REEs took nearly 150 years, as shown in Fig. 1, which illustrates the timeline of REE discoveries and separations.

Currently, there is no unified standard for classifying the 17 REEs. However, the most common method categorizes them based on electronic structure into light rare earth elements (LREEs) and heavy rare earth elements (HREEs). LREEs include lanthanum (La), cerium (Ce),

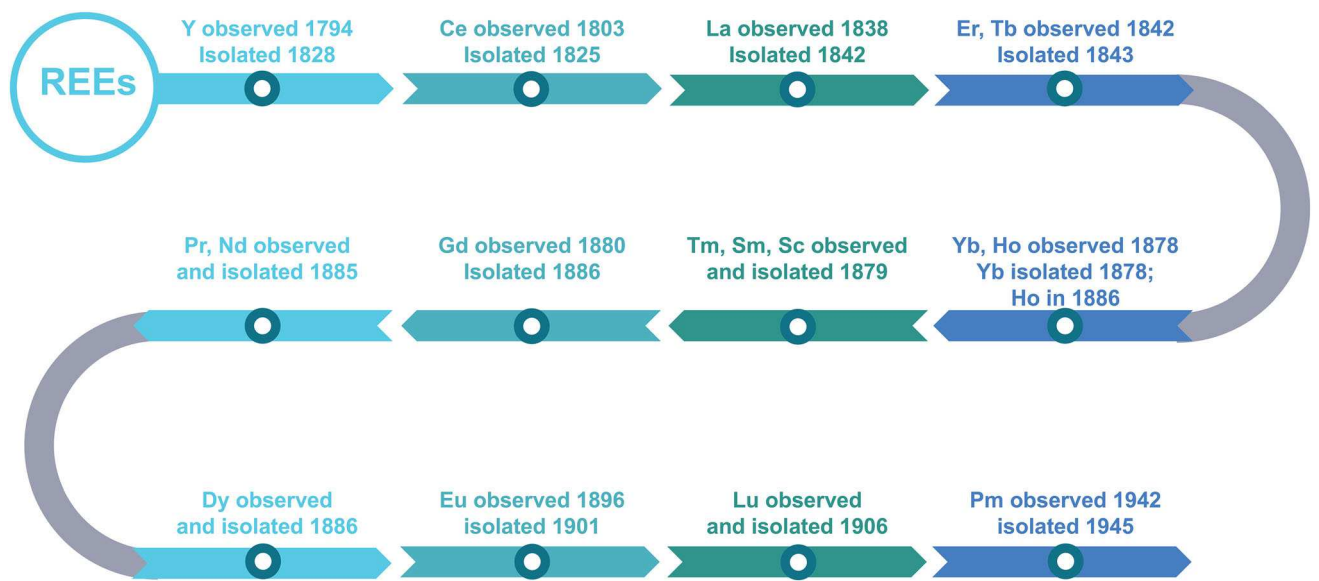


Fig. 1 Timeline of the discovery and separation of rare earth elements (data from Refs. [19, 31, 32])

Table 1 Symbols, oxides, types, and abundances of rare earth elements [34, 36]

Atomic number	Elements	Symbol	Oxide	Type	Abundance (ppm)
21	Scandium	Sc	Sc ₂ O ₃		14
39	Yttrium	Y	Y ₂ O ₃	Heavy	21
57	Lanthanum	La	La ₂ O ₃	Light	31
58	Cerium	Ce	Ce ₂ O ₃	Light	63
59	Praseodymium	Pr	Pr ₂ O ₃	Light	7.1
60	Neodymium	Nd	Nd ₂ O ₃	Light	27
61	Promethium	Pm	–	Light	–
62	Samarium	Sm	Sm ₂ O ₃	Light	4.7
63	Europium	Eu	Eu ₂ O ₃	Light	1
64	Gadolinium	Gd	Gd ₂ O ₃	Light	4
65	Terbium	Tb	Tb ₂ O ₃	Heavy	0.7
66	Dysprosium	Dy	Dy ₂ O ₃	Heavy	3.9
67	Holmium	Ho	Ho ₂ O ₃	Heavy	0.83
68	Erbium	Er	Er ₂ O ₃	Heavy	2.3
69	Thulium	Tm	Tm ₂ O ₃	Heavy	0.3
70	Ytterbium	Yb	Yb ₂ O ₃	Heavy	2
71	Lutetium	Lu	Lu ₂ O ₃	Heavy	0.31

praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), and gadolinium (Gd), while HREEs include terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), and yttrium (Y) [19]. Scandium (Sc), due to its lack of close association with other REEs in natural minerals, is typically classified as neither LREE nor HREE [19, 33, 34].

REEs are not truly rare, despite the misleading name. They are quite abundant in the Earth's crust [14], with the sum of overall abundances ranging from 150 to 220 ppm, surpassing some common metals such as copper (55 ppm) and zinc (70 ppm) [7, 35]. The symbols, oxides, types, and abundances of REEs are shown in Table 1 [34, 36].

Additionally, the abundance of REEs exhibits two prominent patterns. Firstly, LREEs generally have high



abundances, such as cerium (63 ppm), lanthanum (31 ppm), and neodymium (27 ppm), with individual REE abundances in the Earth's crust exceeding some commonly used metals like lithium (20 ppm) and cobalt (20 ppm). In contrast, some REEs, like thulium (0.3 ppm) and lutetium (0.31 ppm), are extremely low in abundance, with differences exceeding 200-fold. Moreover, HREEs, except for dysprosium, erbium, and yttrium, have abundances below 1 ppm in the Earth's crust [4, 34, 36–38]. Notably, promethium is the only REE not naturally occurring, with 38 isotopes having half-lives ranging from 1 microsecond to 17.7 years, primarily sourced from radioactive decay [26, 39, 40]. Secondly, REEs with even atomic numbers are more common than those with odd atomic numbers, resulting in higher abundances of even-numbered REEs [4, 41].

Furthermore, REEs mostly exist in trivalent oxidation states, but Eu and Ce differ slightly, with Eu existing in a divalent state and Ce in a tetravalent state [36, 41].

2.2 Applications

REEs are highly important and commercially valuable resources. Their unique physical and chemical properties and broad applications make them increasingly significant and strategic. REEs are closely linked to social and economic development, scientific and technological advancements, energy storage, and environmental protection. They have profound impacts on over 40 industries across more

than 10 fields, including aviation, telecommunications, electronics, energy, transportation, military, and medical equipment [4, 42–45]. Accordingly, the REE market is typically divided into nine sectors: catalysts, polishing, glass, phosphors and pigments, metallurgy, batteries, magnets, ceramics, and others [19]. Table 2 shows the applications of various REEs.

REEs are widely utilized in various materials due to their excellent optical, catalytic, magnetic, and electrical properties [19]. Even small additions of REEs can significantly enhance material performance, making them stronger, lighter, or more conductive [35, 36]. Their utilization in portable electronic devices, permanent magnets, hybrid and electric vehicles, wind turbines, fluorescent lamps, and solar cells makes these devices more powerful, efficient, durable, compact, and reliable [13, 34, 48–50]. Moreover, REE permanent magnets and some REE alloys are crucial for many military weapons systems, and indispensable for national defense and security [49, 51].

Overall, REEs are considered strategic commodities that ensure national economic prosperity and security, attracting the attention and competition of many countries [51, 52]. Due to their excellent properties, REEs are known as “seasoning metals” in Germany, “industrial MSG (monosodium glutamate)” in China, and “vitamins of modern industry” in Japan [35]. As shown in Table 2, LREEs such as lanthanum, cerium, praseodymium, and neodymium have the widest applications, covering almost all fields requiring REEs as raw materials. The rarest

Table 2 Applications of various rare earth elements [17, 36, 46, 47]

	Sc	Y	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Aerospace	✓	✓			✓	✓					✓						
Automobile		✓	✓	✓	✓	✓				✓	✓	✓					
Battery		✓	✓	✓	✓	✓	✓	✓		✓	✓		✓	✓			✓
Chemical processing	✓		✓	✓	✓	✓		✓									✓
Glass	✓		✓	✓	✓	✓			✓	✓	✓		✓	✓	✓	✓	
Lamp	✓	✓	✓	✓	✓		✓		✓		✓	✓	✓		✓	✓	✓
Magnet		✓	✓	✓	✓	✓		✓		✓		✓	✓				
Medical	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓		✓		✓	✓	
Nuclear industry		✓						✓	✓	✓		✓		✓			
Oil refining	✓		✓	✓	✓	✓											✓
Semiconductor	✓		✓	✓	✓	✓				✓			✓	✓			✓
Flat display		✓	✓	✓					✓		✓						
Laser		✓	✓	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓	✓	✓
Ceramics		✓	✓	✓	✓	✓											
Phosphors		✓	✓	✓					✓	✓	✓						
Metal alloys			✓	✓	✓	✓											
Polishing compounds			✓	✓	✓												

HREEs are employed in a few highly specialized high-tech fields, such as efficient lighting applications, renewable energy technologies, and permanent magnets [6, 34]. This table provides a useful reference for understanding the diverse applications of REEs in various industries.

Notably, REE-based permanent magnets are indispensable in the new energy sector, such as portable electronics, electric vehicles, and wind turbines. Due to their extensive application in the new energy industry, REEs are called “green elements”. Especially in the context of carbon peak and carbon neutrality goals, there is growing emphasis on environmental protection. Countries are turning to clean energy, leading to continued global demand for REEs [34, 53, 54]. Critically, dysprosium, terbium, yttrium, europium, and neodymium are key raw materials for emerging energy technologies, with significant demand. Among these five highly demanded REEs, except for neodymium, the other four are rare HREEs [7]. However, except for China, other countries have almost no production capacity for HREEs. Therefore, the scarcity of HREE reserves is even more crucial [34, 55].

Although REEs have permeated most aspects of society, their application path has been long and winding from a historical perspective. Since the first discovery of REEs in 1788, they have undergone centuries of research and application. Until the 1950s, global annual production and consumption were less than 5,000 metric tons of rare earth oxides (REO) equivalent. Since the 1960s, with the development of fields such as television screens, the petroleum industry, and computer systems, REEs began to be widely employed, though still limited. Entering the twenty-first century, with the rise of new energy and high-tech industries, the application range and market demand for REEs have continuously expanded, fully reflecting their value [19, 32].

In particular, under the current context of deeply ingrained low-carbon concepts, REEs play a key role in

decarbonization-related fields such as electric vehicles and wind power. Global commitments from these applications are pledged to be 321 Gt CO₂-equivalent [56]. Therefore, it is foreseeable that the demand for REEs will remain very important and substantial in the future.

3 Resources and production

3.1 Global resources and production

The global reserves of REEs amount to 110 million tons and are concentrated in a few countries and regions. China, Vietnam, Brazil, Russia, India, Australia, the USA, and Greenland are the major countries and regions with significant REE reserves. Among them, China has 44 million tons, accounting for 38.1% of the global total [57], which is the highest in the world. The distribution of REEs worldwide is shown in Fig. 2. However, the high concentration of REE deposits in limited geographies is unfavorable to the global industry, as production can be easily influenced by market manipulation, natural disasters, political changes, or environmental issues [49].

Currently, REEs have been discovered in over 250 types of minerals worldwide [36]. REE deposits can be classified into primary and secondary deposits, with primary deposits being categorized as carbonatite, alkaline granite, phosphate rock, and iron oxide–copper–gold types [58, 59]. Secondary deposits can be classified as ion-adsorption (also known as weathered crust) type, monazite, xenotime sand, beach sand, and deep-sea mud types [13, 26]. Carbonatite and sand deposits have always been the major sources of LREEs worldwide, while ion-adsorption clay deposits are the main sources of HREEs [38]. Among the known 250 REE minerals, the main minerals are bastnäsite, monazite, ion-adsorption clay, and xenotime, which store about 95% of the total REEs [60]. At present, only three main REE mineral types (bastnäsite, monazite, and xenotime) are commercially exploited [61].

There are over 1000 REE deposits worldwide [50], but only a few are currently being exploited. The major mines in operation are Bayan Obo in China, Mountain Pass in the USA, and Mount Weld in Australia [35, 36, 61]. It is worth noting that 70 wt% of the world's REEs are extracted from bastnaesite [62].

Presently, about 51.4% of the world's REO resources are from carbonatite deposits [16]. These carbonatite deposits are mainly distributed in countries such as China, Brazil, Australia, and Canada, with 42% of REE deposits coming from medium-sized mines [19]. Moreover, the world's three major light REE mines (Bayan Obo in China, Mountain Pass in the USA, and Maoniuping in China) are all carbonatite types [26].

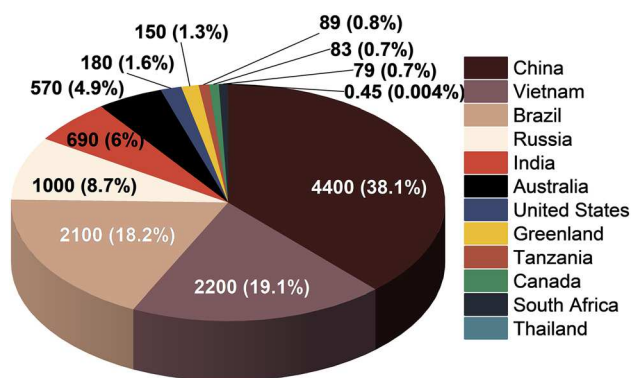


Fig. 2 REE reserves (data in terms of ten thousand metric tons REO equivalent) of major countries and regions (2023) (data from Ref. [57])

Table 3 Some major rare earth minerals and their chemical composition [6]

Mineral Name	Chemical formula
Allanite	(REE, Ca, Y) ₂ (Al, Fe ³⁺) ₃ (SiO ₄) ₃ (OH)
Ancylite	Sr(REE)(CO ₃) ₂ (OH)·H ₂ O
Apatite	(Ca, La) ₅ (PO ₄) ₃ (F, Cl, OH)
Bastnaesite	(REE)CO ₃ F
Brannerite	(U, Ca, Y, REE) (Ti, Fe) ₂ O ₆
Britholite	(REE, Ca, Th) ₅ (SiO ₄ , PO ₄) ₃ (OH, F)
Burbankite	(Na, Ca) ₃ (Sr, Ba, Ce) ₃ (CO ₃) ₅
Cerianite-(Ce)	(Ce ⁴⁺ , Th)O ₂
Eudialyte	Na ₄ (Ca, REE) ₂ (Fe ²⁺ , Mn ²⁺ , Y)ZrSi ₈ O ₂₂ (OH, Cl) ₂
Fergusonite	(La, Y)NbO ₄
Fluorapatite	(Ca, REE, Na) ₅ (PO ₄) ₃ (F, OH)
Gadolinite	(REE, Y) ₂ Fe ²⁺ Be ₂ Si ₂ O ₁₀
Gittinsite	CaZrSi ₂ O ₇
Gorceixite	(Ba, REE)Al ₃ (PO ₄) ₂ (OH ₅ ·H ₂ O)
Limorite-(Y)	Y ₂ SiO ₄ CO ₃
Kainosite	Ca ₂ (Y, REE) ₂ Si ₄ O ₁₂ CO ₃ ·H ₂ O
Loparite-(Ce)	(Na, Ce, La, Ca, Sr)(Ti, Nb)O ₃
Loparite	(La, Na, Ca)(Ti, Nb)O ₃
Monazite	(REE, Th)PO ₄
Mosandrite	(Na, Ca) ₃ Ca ₃ La (Ti, Nb, Zr)(Si ₂ O ₇) ₂ (O, OH, F) ₄
Parisite	Ca(REE) ₂ (CO ₃) ₃ F ₂
Pyrochlore	(Ca, Na, La) ₂ Nb ₂ O ₆ (OH, F)
Rhabdophane	(REE)PO ₄ ·H ₂ O
Rinkite (Rinkolite)	(Ca, La) ₄ Na(Na, Ca) ₂ Ti(Si ₂ O ₇) ₂ (O, F) ₂
Steenstrupine	Na ₁₄ La ₆ Mn ₂ Fe ₂ (Zr, Th)(Si ₆ O ₁₈) ₂ (PO ₄) ₇ ·3H ₂ O
Synchysite	Ca(REE)(CO ₃) ₂ F
Thalenite-(Y)	Y ₃ Si ₃ O ₁₀ OH
Xenotime	(Y, Gd, Dy, Er)PO ₄
Zircon	(Zr, La)SiO ₄

Monazite is the second-largest major type of REE deposit [36]. It is important to note that monazite is a REEs-Th phosphate mineral that is also applied to extract light REEs. Thorium is radioactive, and its handling is complex, dangerous, and expensive, which has raised concerns. As a comparison, Bastnäsite has relatively low thorium content and is mainly composed of La, Ce, and Nd [9].

Table 3 lists some of the main REE minerals and their chemical compositions. In 2023, global REE mining output increased to 350,000 metric tons of REO equivalent, which was about 3.18 times that of 2011 (110,000 metric tons of REO equivalent). Among the major REE producers (China, the USA, Australia, and Myanmar, accounting for 96.86% of the world's total production), China has the highest

production, with 240,000 metric tons, accounting for 80%. Compared to 2021 and 2022, both China's REE production and its share in the global market have significantly increased [6, 36, 58, 59, 63–70]. The main countries and regions with REE reserve and production are shown in Fig. 3. REE production has been steadily increasing year by year. In addition, it is worth noting that Myanmar's REE production has rapidly increased from 12,000 tons in 2022 to 38,000 tons in 2023, more than tripling. This significant change warrants attention. According to the study of Wang et al. [32], global REE production will continue to increase, with a minimum predicted peak of 1.28 million metric tons in 2115 and a maximum predicted peak of 40 million metric tons in 2118.

Figure 4 shows the global total demand for REEs, excluding China, from 2010 to 2025. As can be seen, except for a decline in demand from 2010 to 2012, there has been a gradual upward trend since 2012 [72]. The continuously growing high demand will inevitably lead to resource shortages, especially for non-renewable resources. In recent years, there has been a shortage of REE resources of more than 10% [73], with a risk of long-term shortage as well [34, 74]. In particular, the large-scale construction of high-efficiency lighting and renewable energy technologies, such as electric vehicles and wind turbines, will result in an increasing demand for REEs [19, 74].

In more specific terms, the predicted demand for REOs in various industries (products) in 2019 and 2025 is shown in Fig. 5 [75].

Compared to 2019, it is expected that by 2025, the demand for REEs in all sectors (products) except for batteries will surge sharply, particularly in the fields of magnets and phosphors, which will increase by more than double. Additionally, the demand for catalysts will also increase significantly. It is evident that these industries will show excellent development momentum in the coming years. Moreover, with the development of strategic emerging industries, the production of main functional REE materials such as magnetic and catalytic materials is expected to grow at a rate of over 15% annually [76].

According to predictions, by 2030, the demand for REEs in clean technology will reach 519,000 tons of REO, with Nd and Dy accounting for 75% and 9%, respectively. This indicates that Nd will have significant impacts on the development of REE new projects and clean technologies in the coming decades. Correspondingly, REEs, such as Nd, are expected to become scarce soon [19]. The REE prices for 2020, 2025 (forecast), and 2030 (forecast) are shown in Table 4. Due to the diversity of REE oxides, there is no unified trend of upward or downward movement among them. Depending on the supply and demand relationship, different elements have shown either an increase or decrease in price. From Table 4, the prices of

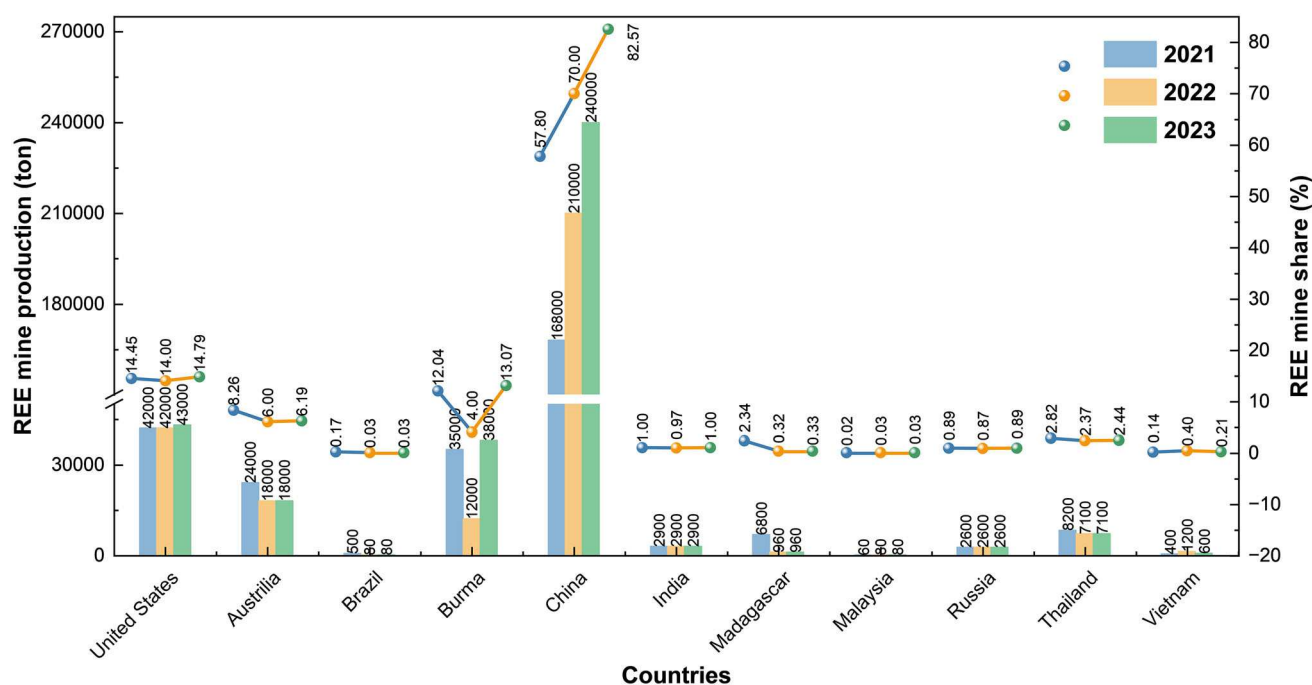


Fig. 3 REE production and percentage (in terms of REO equivalent) of major countries in 2021, 2022, and 2023 (data from Refs. [57, 71])

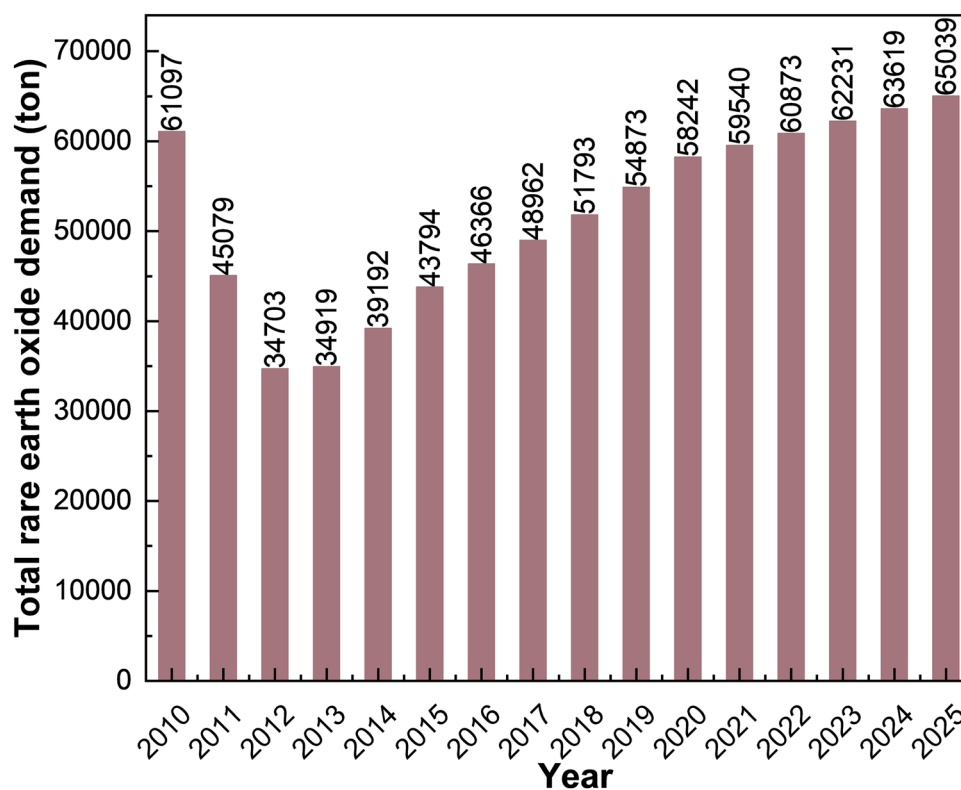


Fig. 4 Total rare earth oxide demand worldwide excluding China from 2010 to 2025 (in metric tons) (data from Ref. [72])

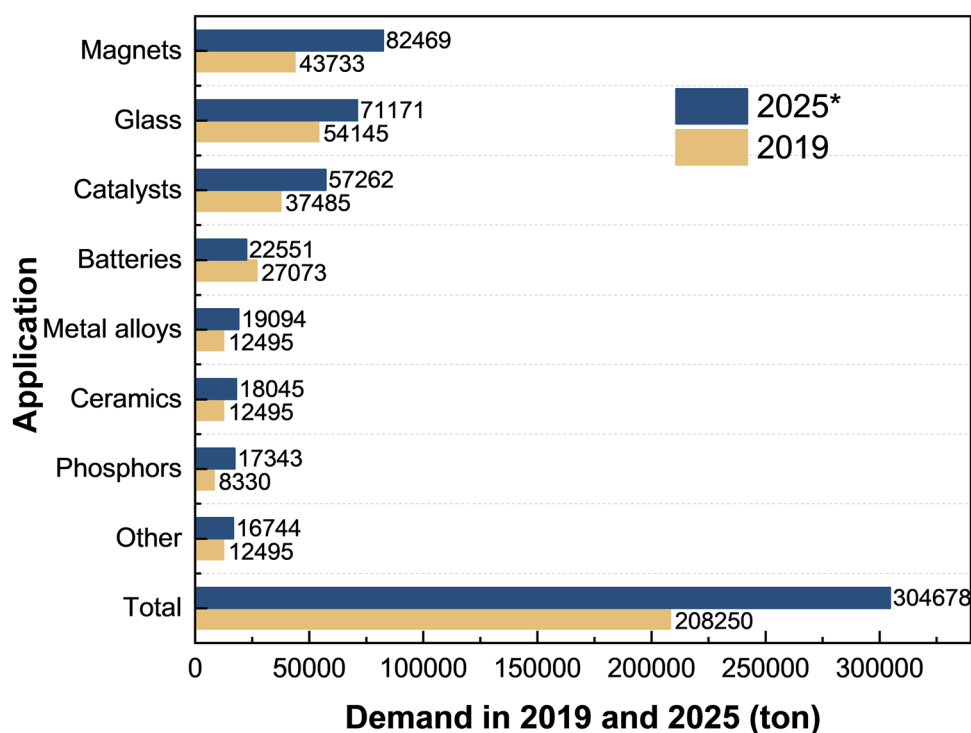


Fig. 5 Demand for rare earth oxides worldwide in 2019 and 2025 by end application (in metric tons) (data from Ref. [75])

Table 4 Rare earth oxide prices worldwide in 2020 and a forecast for 2021 to 2030 [77]

Rare earth oxide prices worldwide in 2020 and a forecast for 2021 to 2030* (US\$ kg ⁻¹)	2020	2025*	2030*
Lanthanum oxide	1,643,000	1,650,000	1,590,000
Cerium oxide	1,663,000	1,660,000	1,610,000
Praseodymium oxide	45,763,000	94,500,000	59,500,000
Neodymium oxide	49,144,000	77,500,000	45,500,000
Samarium oxide	1,756,000	1,620,000	1,520,000
Europium oxide	30	28	28
Gadolinium oxide	24,678,882	25,468,252	27,287,413
Terbium oxide	671	850	700
Dysprosium oxide	260	530	440
Holmium oxide	402.10	290	250
Erbium oxide	23	18	16
Ytterbium oxide	14.70	14.55	14.55
Lutetium oxide	621.57	625.79	640.34
Yttrium oxide	2.94	2.75	2.60
Scandium oxide	938.92	91.86	916.85

* Represents the forecast prices of rare earth oxides

neodymium oxide and praseodymium oxide are expected to increase significantly in the short term (2020–2025) but are expected to decline slightly by 2030, gadolinium oxide

is expected to maintain a growth trend between 2020 and 2030 [77].

3.2 China's resources and production

China, as the world's largest producer of REEs, holds the greatest reserves, production capacity, and export volume [78]. For decades, China has dominated the global REE market, at one point contributing 97% of the world's total production [79].

China's REE mineral resources are widely distributed, primarily concentrated in the Inner Mongolia Autonomous Region, five southern provinces (Jiangxi, Guangdong, Fujian, Guangxi, and Hunan), as well as Shandong, Sichuan, and Yunnan [67, 80, 81]. The country's major REE sources include the Bayan Obo mine in Inner Mongolia, bastnaesite deposits in Sichuan and Shandong, and ion-adsorption mines in southern China [42]. Notably, the Bayan Obo mine in Inner Mongolia and the Maoniuping mine in Sichuan are ranked first and third among the world's largest REE mines, respectively [26]. In fact, over half of China's REE production comes from the Bayan Obo mine, the largest REE mine globally [80].

China's distinct advantage lies in its HREE resources, which account for more than 80% of the world's HREE reserves [82]. The southern ion-adsorption mines are the world's primary source of HREEs [80], containing large quantities of critical elements like terbium (Tb),

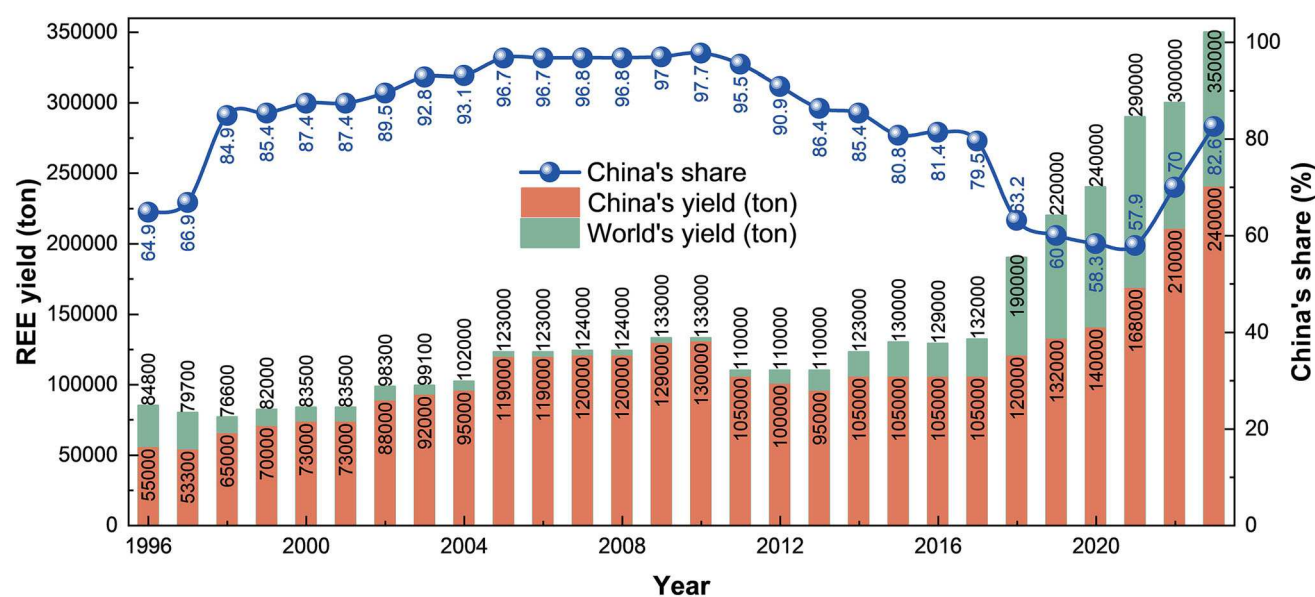


Fig. 6 China's annual rare earth production and its global share over the years (data from Ref. [71])

dysprosium (Dy), erbium (Er), and yttrium (Y), in concentrations more than ten times higher than those found in bastnaesite and monazite [82]. In these deposits, HREEs are adsorbed by clay minerals in ionic form [83]. Comparing the proportion of ion-adsorption HREE resources in each southern province of China, with Jiangxi and Guangdong holding particularly significant roles, contributing 36% and 33% of the mining output, respectively [84].

In addition to leading the world in production, China has also held the position of the world's largest REEs producer for many consecutive years [85].

According to data released by the USGS, from 2003 to 2023, China's REEs production accounted for over 90% of global production, and in 2010, China's REEs production reached as high as 97.74%, almost single-handedly supplying the world [71]. As a result, the world has become dependent on China, and many chemical, steel, and non-ferrous metal industries in the USA, as well as their downstream customers, rely heavily on imports from China, as some raw materials can only be sourced from China [86]. The European Union relies on China for 98% of its REEs imports [87]. Japan imports 82% of its REEs from China, which accounts for 40% of China's total REE exports [88]. The production of REEs in China and their global share are shown in Fig. 6.

Overall, China's REE production has shown a growth trend over the past years. During the period from 2000 to 2020, the production of LREEs in China has increased rapidly, while the production of HREEs has decreased. In 2020, the production of LREEs was 121,000 tons, and the HREEs were 19,000 tons. Compared with 2000, the production of LREEs increased by 78.2%, while the

production of HREEs decreased by 38.4%. In 2020, the production of LREEs such as La, Ce, Pr, and Nd increased by 75.5% to 118.9% compared with the year of 2000, reaching 35,000, 59,000, 9,000, and 25,000 tons, respectively. However, the production of HREEs such as Gd, Dy, and Y decreased by 34.8% to 43.9% from the year of 2005 [42].

In 2018, the output value of China's REE industry chain was about 90 billion CNY (~ 1.36 billion in USD), of which REE functional materials accounted for 56%, with an output value of about 50 billion CNY, and metallurgy and separation accounted for 27%, with an output value of about 25 billion CNY. Among them, REE permanent magnet materials have the highest proportion in REE functional materials, accounting for 75% with an output value of about 37.5 billion CNY, and catalytic materials account for 20%, with an output value of about 10 billion CNY [3]. Figure 7 shows the prices of various REOs in January 2023 in China [89]. Due to their scarcity, the prices of heavy REOs are generally higher than those of light REOs. Among them, holmium oxide is the most expensive, reaching 935,000 CNY per metric ton, while europium oxide is the least expensive, only at 195 CNY per metric ton. The price difference between the two can reach thousands of times, which is closely related to the application field and the supply-demand relationship.

In 1988, China's REE production exceeded that of the USA, making it the world's top REEs producer [3]. In the mid-1990s, China surpassed the USA to become the world's largest REEs supplier and has since maintained its dominant position [43, 80]. China not only ranks first in terms of reserves and production, but also has an absolute advantage in all aspects of REE production to application.

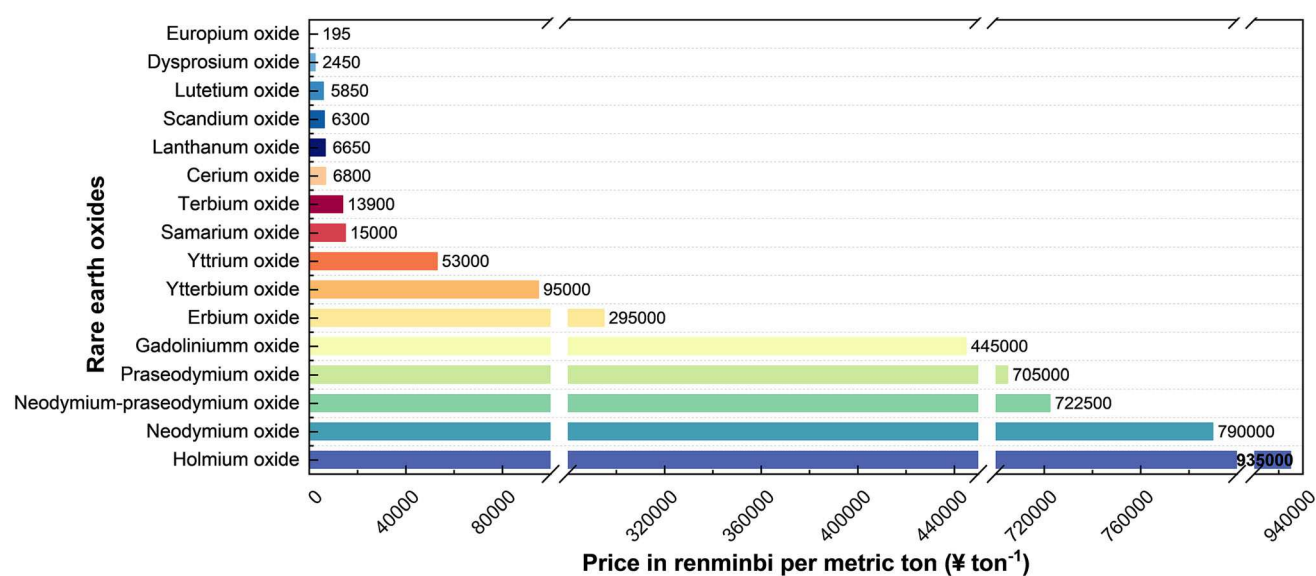


Fig. 7 Average Chinese domestic prices for selected rare earth oxides as of January 2023 (in renminbi per metric ton) (data from Ref. [89])

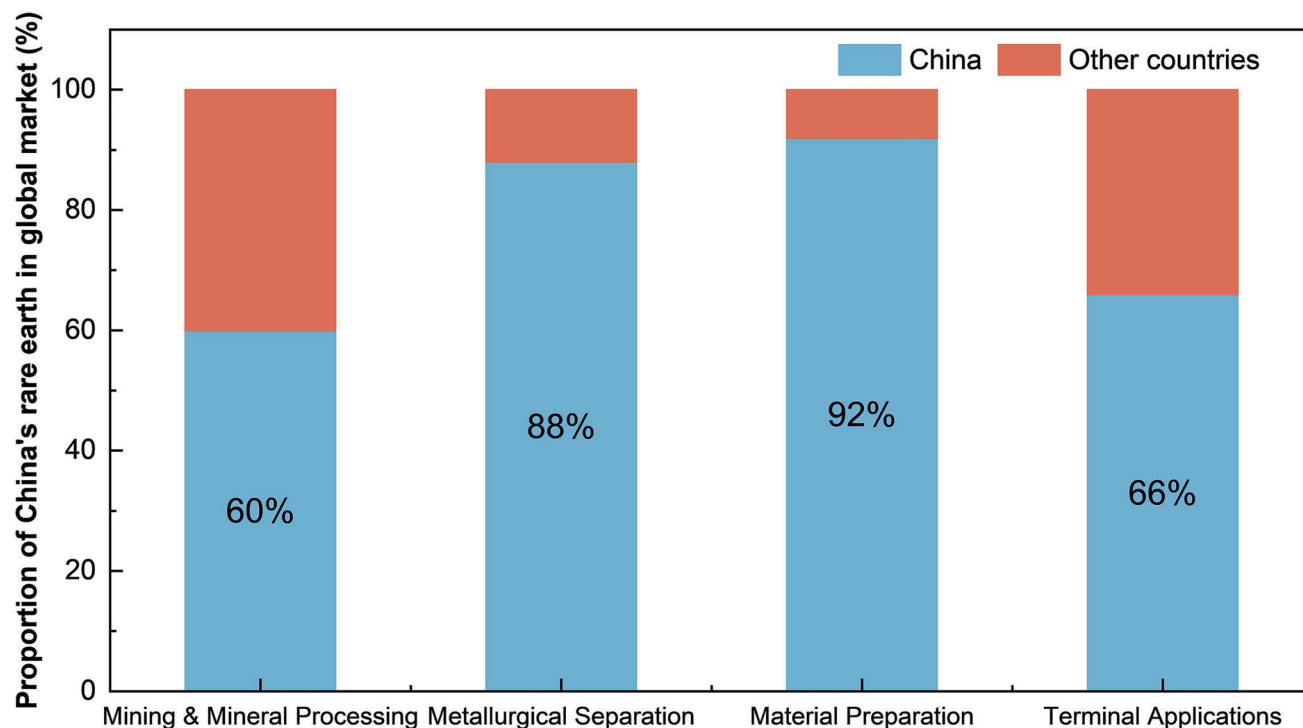


Fig. 8 Proportion of China's rare earth industry chain in each stage's global market (data from Ref. [79])

As shown in Fig. 8, China ranks first in the world in mining and mineral processing, metallurgy and separation, material preparation, and terminal applications (all exceeding 60%), especially in the fields of metallurgy and separation and material preparation, which has reached about 90%, occupying an absolutely dominant position in the world [79].

According to another source in more detail [79], in the REEs mining and beneficiation sector, China surpassed the USA for the first time in 1986 and has since maintained its dominant position. In 2021, the global production of REE mineral products was about 280,000 tons, of which China's production accounted for 60%. In the metallurgical and REE separation sector, China created the cascade

extraction method in the 1980s and it was widely applied, which led to China's REO production surpassing the USA to become the world's largest in the 1990s, and even resulting in China's REE production accounting for more than 97% of the world's total in 2010. Although the proportion has declined in recent years, for example, being 88% in 2020, China is still the most important producer in the global REE market [79]. Specifically, almost all the HREEs can only be processed in China [79]. In the area of REE functional material preparation, China not only controls the raw materials but also controls the production of many key intermediates entering many high-tech emerging industries [86].

In addition, China has attracted substantial foreign investment to establish factories through its resource and industry advantages, as well as related policies. China has quickly formed permanent magnet industry clusters primarily in Ningbo, the Beijing-Tianjin region, Baotou, and Ganzhou. This has resulted in China's share of REE permanent magnet materials production being 92%, nearly monopolizing the industry. Furthermore, China produces about 70% of the world's REE hydrogen storage materials and 45% of the world's REE catalytic materials. However, there is still a gap between China's high-end REE materials and those of Western countries. According to statistics, China's production of high-end REE permanent magnets accounts for only 23% of the global market share, while other countries that produce high-end REE permanent magnets include Japan (30%–40%), the USA (10%–15%), and Germany (5%–10%) [43, 86].

In terms of application, China has been the world's largest REE consumer since 2002. In 2020, China consumed 115,000 tons of REEs, with REE permanent magnets being the largest consumption area, accounting for as much as 42% [79].

In summary, although China is a major producer and consumer of REEs, in some areas, its relevant technology still lags behind that of developed countries, and there is still a certain gap between China and the world's high-end leading REE countries [86].

4 Mining and recycling

From a sustainable development perspective, understanding the lifecycle of REEs is crucial because it helps us gain a comprehensive understanding of how these resources are utilized. The lifecycle of REEs encompasses various stages such as mining, metallurgy, circulation, and recycling. These stages are interconnected and hold the potential for cross-fertilization. By viewing the REEs lifecycle as sustainable and circular, we can maximize resource utilization and transform waste into wealth and secondary resources.

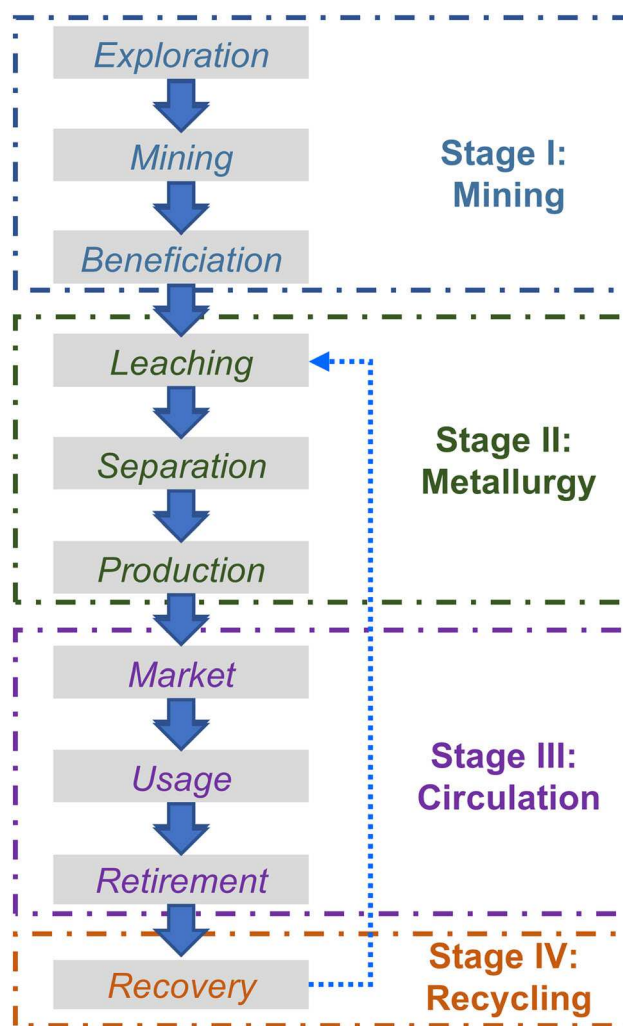


Fig. 9 The life cycle of rare earth elements

As shown in Fig. 9, the lifecycle of REEs consists of four main stages: mining, metallurgy, circulation, and recycling.

In fact, the mining process typically includes metallurgical processes, as some REE minerals require in situ leaching, making it difficult to completely separate mining and metallurgy in the REE field. Therefore, we have included the metallurgical process in the mining section for discussion. Additionally, while recovery and recycling are essentially similar, there are subtle differences. Recovery focuses primarily on extracting specific REEs from secondary resources for reuse, which is the key method to achieve recycling goals. Hence, in this section, we categorize the technologies into mining and recycling for detailed discussion.

4.1 Mining

In general, the mining process involves first identifying the REE ore through exploration, then mining it using underground or open-pit methods, followed by crushing and

grinding for size reduction and classification. Finally, the material is enriched through techniques such as flotation, gravity separation, and magnetic separation, which can greatly improve efficiency in subsequent metallurgical processes [17, 46]. It is worth noting that ion-adsorption type REE ore requires unique processing techniques since traditional beneficiation methods are ineffective for this type of ore, and they usually do not include beneficiation steps [90, 91].

The objective of the REE metallurgical stage is to separate REE cations from the ore and produce refined REO products. While there is a diversity of REE minerals, the major metallurgical process for all of them primarily relies on hydrometallurgical methods. This involves leaching REE cations into an aqueous solution, followed by the individual separation of various REEs, ultimately resulting in the production of REO products. Although different minerals may employ varying methods, the fundamental processes and goals are similar. Leaching methods encompass acid leaching, acid-oxidant co-leaching, alkaline leaching, and acid-alkali joint leaching. In this regard, it is important to note that the processes of REEs purification and separation share a fundamental similarity. Purification involves separating REEs from other elements in the aqueous phase, while separation focuses on isolating different REEs from each other. Ultimately, both are selective processes relying on the exploitation of the differences in their chemical properties to achieve the sequential separation of individual ions. The primary methods for REEs purification and separation include solvent extraction, ion exchange, adsorption, precipitation, membrane separation, etc. Subsequently, further steps involve methods such as evaporative concentration or precipitation to obtain solid REE compounds, followed by additional refining steps to obtain the final REE products [7, 15, 33, 82, 92].

Particularly, in the context of REEs mining and metallurgy, it is of paramount importance to reduce environmental pollution, lower production costs, and enhance production efficiency. Future technologies should prioritize these aspects while aligning with sustainable and low-carbon development principles to remain competitive and hold industrial prospects. Below, this paper will categorically review research progress concerning different types of REE minerals.

4.1.1 Ion-adsorption type REE ores

In ion-adsorption type REE ore, REEs are typically present in the form of ion-adsorption on the surfaces of clay minerals such as kaolinite, feldspar, and mica. The content of REEs in this mineral is extremely low, ranging from 0.03 wt% to 0.3 wt% [93], and traditional beneficiation

techniques such as gravity separation and flotation are ineffective in concentrating REEs from such a resource type [90, 91]. Therefore, unconventional methods are required for processing ion-adsorption type REE ore.

For ion-adsorption type REE ore, an effective method involves leaching and ion exchange using cations with specific properties. Commonly used leaching agents include magnesium salts, calcium salts, iron salts, and ammonium salts. In general, cations with smaller ionic radii and higher valence states have stronger leaching capabilities for REEs [90]. The leaching capability sequence is as follows [94, 95]: $\text{Al}^{3+}/\text{Fe}^{3+} > \text{Mg}^{2+}/\text{Fe}^{2+} > \text{Ca}^{2+} > \text{NH}_4^+ > \text{Na}^+$.

Initially, inexpensive and readily available sodium chloride was employed as a leaching agent. However, this method required high concentrations of sodium chloride, which led to soil salinization and environmental damage. Additionally, sodium salt had relatively weak leaching capabilities, gradually leading to discontinuation of this method [93, 95, 96].

Ammonium salts, are more effective leaching agents compared to sodium salts [90, 97]. Among them, ammonium bisulfate is the most widely utilized ammonium salt leaching agent [98]. They exhibit stronger leaching ability, and the advantage of using ammonium salts is that satisfactory results can be achieved with low-concentration solutions, reducing chemical consumption. However, the utilization of ammonium salts generates a large amount of ammonium nitrogen wastewater, which poses serious environmental challenges [90, 95, 96, 99, 100].

Considering sustainability, there is a need to develop alternative processes that do not produce ammonium nitrogen wastewater. Extensive research has shown that magnesium salts are a rational alternative to ammonium salts [99, 101]. This method avoids ammonium nitrogen wastewater pollution from the source while achieving ideal leaching rates [102]. Furthermore, due to the thermodynamic differences in the chloride ion dissolution between magnesium salts and ammonium salts, the impurity removal rate was improved by more than 13%. As a result, the magnesium salt leaching method has gained wide acceptance [94]. However, magnesium salts employed as leaching agents can lead to soil salinization and compaction due to the accumulation of Mg^{2+} in the soil. Similarly, the application of other metal salts may also result in the buildup of corresponding cations [103, 104]. In addition, the commonly applied magnesium salt leaching agent, magnesium sulfate, has a weaker leaching ability compared to ammonium sulfate, which leads to higher reagent consumption during the leaching process [105].

Combinations of multiple salts have also been utilized for leaching ion-adsorption type REE ore. For example, using calcium salts and magnesium salts as leaching agents

while maintaining a calcium-magnesium mass ratio between 8 and 12 can meet soil nutrient ratio requirements, which is beneficial for the environment. Moreover, using two different magnesium salts as leaching agents has been studied, and it was found that when the molar ratio of magnesium chloride to magnesium nitrate is 4:6, the leaching level can reach 95.22% [106]. For ion-adsorption type REE ores, the most commonly selected leaching agents are ammonium sulfate and magnesium sulfate [97].

However, the extraction of ion-adsorption type REE ores primarily relies on ion exchange between the cations in the leaching agent and the REE ions adsorbed on the surface of clay minerals. This reaction is reversible: during in situ leaching, the REE ions displaced by the leaching agent may be re-adsorbed by the surrounding clay minerals, resulting in a reduced extraction efficiency. To overcome this issue, an excess amount of leaching agent is typically added to shift the equilibrium between adsorption and desorption, thereby improving REE recovery efficiency [104]. Therefore, in addition to the salt-based leaching solutions mentioned above, acids can also be applied in the leaching process or as auxiliaries to enhance the leaching of ion-adsorption type REE ores [104, 105]. This is because the H^+ ions in acids can participate in ion exchange with REEs in the exchangeable phase, and the complexation effect of organic acid anions can drive the leaching reaction forward. Different organic acids exhibit distinct leaching mechanisms: For instance, acetic acid primarily promotes leaching through ion exchange and acid dissolution, while malic acid and citric acid rely mainly on complexation during the dissolution process [107, 108].

For example, a study by Zhao et al. [104] found that citric acid achieved a higher leaching efficiency at lower concentrations compared to ammonium sulfate and magnesium sulfate. With proper application, citric acid can result in leaching rates as high as 96% across all REEs, and up to 99% specifically for light REEs. This is because organic acids are generally more effective in leaching light REEs [107].

In addition, mixed acid-salt leaching solutions can also be utilized for REEs leaching. For instance, Lai et al. [105] employed a combination of magnesium sulfate and ascorbic acid as the leaching agent. Ascorbic acid can form stable complexes with REE ions, thereby promoting the ion exchange reaction. The study showed that under conditions of pH 2.00, magnesium sulfate 0.15 mol L^{-1} , and ascorbic acid 1.0 g L^{-1} , the REE leaching efficiency reached 107.5% [105].

Beyond conventional chemical reagents, bioleaching is also considered a promising approach due to its environmentally friendly and sustainable nature, along with mild reaction conditions [109]. For example, Meng et al. [98] used citrate salts synthesized from microbial metabolites—

(Na)₃Cit—as a primary leaching agent, achieving a high leaching efficiency of 98% for ion-adsorption type REE ores. Notably, this study was conducted by a heap leaching method.

Zhou et al. [2] applied an *Aspergillus niger* strain to leach ion-adsorption type REE ores through three different methods. The study found that the spent medium method achieved the highest leaching efficiency (102%), compared to the one-step method (84%) and the two-step method (93%). The main organic acids produced by the *Aspergillus niger* strain were gluconic acid (23 mg mL^{-1}), citric acid (14 mg mL^{-1}), and succinic acid (5 mg mL^{-1}). However, this approach is relatively costly due to the production of the required microbial metabolites. Therefore, future research should focus on identifying low-energy-consuming microorganisms and utilizing more cost-effective materials to produce the nutrients needed for microbial growth.

For the leaching methods of ion-adsorption REE ore, initially, pond leaching was utilized. This method relied on labor to strip REE minerals and transfer them to leaching ponds for extraction. However, the stripping process caused damage to the terrain and vegetation, generated a large number of tailings, and involved high manual labor intensity. It had low resource utilization efficiency, low overall efficiency, and small production capacity, making it unsuitable for continuous operation. Consequently, this method has been completely phased out [93, 96].

Subsequently, heap leaching technology was developed. In this process, REE ores are stacked in cement leaching tanks with a certain inclination angle at the bottom. During leaching reactions, the leaching agent percolates downward through the ore layers. This method increased production and reduced the consumption of leaching agents. However, the recovery rate for REEs remained low, with an overall REEs recovery rate of less than 70%. Despite these improvements, heap leaching had disadvantages such as high energy consumption and high labor intensity. Additionally, ion-adsorption REE ore often contains radioactive elements like radium, thorium, and uranium, which end up in the acidic leach residue and are challenging to handle. As a result, production facilities were required to construct dedicated radiation-proof warehouses for storing these radioactive residues. Furthermore, there are no alternative solutions for managing these residues, posing serious safety hazards. Consequently, heap leaching technology struggles to meet the requirements of sustainable development [93, 96, 110, 111].

In 1991, the in situ leaching technology for ion-adsorption REE ore was proposed. This method involved dividing the mining area based on geological landmarks and the unified groundwater flow system. Precise mining was conducted within these delineated areas. The

technology utilized injection systems and mother liquor collection systems to establish a comprehensive liquid sealing strategy for ion-adsorption REE ore bodies. This approach significantly increased the recovery of REEs while reducing environmental pollution. The in situ leaching method brought several advantages, including reduced labor intensity, lower infrastructure investments, cost savings, enhanced REE resource utilization efficiency, and a reduction in environmental pollution. It gained widespread recognition and has been continuously applied since its inception. However, it is important to note that in situ leaching is not without its challenges. Improper injection of leaching agents can lead to issues such as landslides, groundwater contamination, and threats to ecological safety in the surrounding environment [46, 50, 93, 96, 102, 112].

After leaching, the obtained REE leachate undergoes purification to produce REE concentrate using ammonium bicarbonate or oxalic acid precipitation. Subsequently, acid leaching and centrifugal extraction separation are applied to purify individual REEs [93, 113, 114]. However, during oxalic acid precipitation, complex salts are formed with impurity ions from the leachate, affecting REE precipitation rates. Currently, despite its challenges, in situ leaching is the primary method for recovering REEs from ion-adsorbed clays, given the advantages mentioned above.

4.1.2 Other REE ores

Unlike ion-adsorption type REE ores, REE minerals such as bastnaesite and monazite typically require beneficiation processes—such as gravity separation, magnetic separation, flotation, or electrostatic separation—prior to metallurgical treatment. The goal of these pretreatments is to reduce the consumption of reagents and energy during subsequent metallurgical processes, thereby improving efficiency. After beneficiation, the REE concentrates usually can reach up to grade of approximately 60%–70%. Among these methods, froth flotation is the most commonly employed technique for mineral enrichment. It utilizes the differences in surface properties (hydrophilicity vs. hydrophobicity) between valuable minerals and gangue to achieve separation. In practice, flotation is often coupled with magnetic or gravity separation methods [115–121].

Compared to ion-adsorption type REE ores, the extraction mechanisms for enriched bastnaesite and monazite are fundamentally different. These minerals rely on the different speciation of REEs under varying pH conditions. By adjusting the pH, REEs can be transformed into soluble forms and recovered in solution. In contrast, ion-adsorption type ores are typically processed via ion exchange [122].

Under acidic conditions, most REEs (such as Ce, Nd, and La) remain stable in aqueous solution as trivalent or

tetravalent ions, whereas, under alkaline conditions, some REEs precipitate as hydroxides (e.g., $\text{Nd}(\text{OH})_3$, $\text{La}(\text{OH})_3$, and $\text{Ce}(\text{OH})_3$) [122]. This principle applies to the leaching mechanisms of bastnaesite, monazite, and their mixture. In acid leaching, REEs react directly with acids to form soluble species. In alkali leaching, REE hydroxide precipitates are first formed and then dissolved in acid to extract REEs. Sometimes, prior to leaching, roasting is adopted to convert REEs into more soluble or reactive forms, thus facilitating subsequent extraction [62, 123].

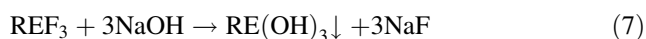
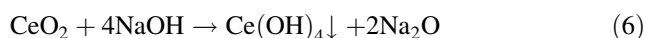
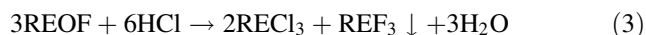
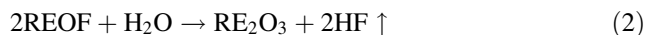
Notably, cerium, a major component of both monazite and bastnaesite, is capable of existing in both +3 and +4 oxidation states. As a result, the recovery of cerium from leach solutions is often more complex than that of other REEs [122].

4.1.2.1 Bastnaesite After the beneficiation process, such as flotation and magnetic separation, the concentrated bastnaesite obtained from the original bastnaesite ore can have a REO grade ranging from 65% to 75% [93] along with 7%–11% fluorine content [124], which can be applied to hydrometallurgical extraction of REEs.

For bastnaesite, direct acid leaching is feasible [122]. However, studies have confirmed that direct leaching with hydrochloric acid results in a leaching efficiency of only 55%–60%, leading to low recovery. Consequently, a subsequent caustic crack leach is often required to recover the remaining REEs from the residue [125].

Currently, the mainstream process for treating bastnaesites is the oxidative roasting-hydrochloric acid leaching process [62]. This method begins with the oxidation roasting of bastnaesite at approximately 700 °C for about 1 h. The purpose of this roasting step is to break down the trivalent REE compounds in bastnaesite into REE oxyfluorides (Reaction (1)), which facilitates the subsequent leaching process. During this process, some of the decomposition products can react with water vapor in the air to form REE oxides (Reaction (2)). In this process, fluorine is released in the form of hydrogen fluoride (HF) gas, which leads to environmental pollution [124]. Next, the trivalent REEs are leached using hydrochloric acid, resulting in soluble REE chlorides and insoluble REE fluorides (Reaction (3)). Most of the cerium is converted to insoluble cerium oxide and cerium fluoride (Reactions (4, 5)). After hydrochloric acid leaching, the cerium oxide and REE fluorides, due to their insolubility in acid, remain in the residue. To handle these residues, an alkali conversion process is employed to transform them into hydroxides that are soluble in acid (Reactions (6, 7)). After removing fluorine by washing it with water, they are then subjected to hydrochloric acid dissolution [124, 126, 127]. Subsequently, through processes like purification, extraction, and precipitation, these trivalent REEs are separated and

purified from the solution, ultimately yielding REE products [96, 128]. The chemical reaction equations are as follows [126]:



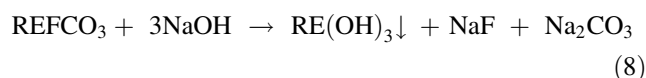
This process has advantages such as low cost and minimal investment. However, it requires special attention when dealing with fluorine. Currently, the primary methods for handling fluorine include: (1) adding calcium, aluminum, or silicon to solidify the fluorine, causing it to be removed in the slag form; (2) using organic solvent extraction to recover fluorine. These methods are employed to effectively manage and recover fluorine during the process.

Cen et al. [62] developed an integrated beneficiation-metallurgy approach for leaching bastnaesite based on an oxidation roasting–hydrochloric acid leaching process. By first performing calcification roasting, followed by gravity separation and then leaching with hydrochloric acid, the recovery rates of REEs, fluorine, and calcium reached 98.07 wt%, 64.76 wt%, and 85.48 wt%, respectively.

In addition, an alkali process has been proposed for bastnaesite. The primary steps in this process involve roasting the bastnaesite with sodium hydroxide. Afterwards, the generated REE hydroxides are washed, filtered, and dissolved in concentrated hydrochloric acid. This results in a REE chloride solution that, after concentration and crystallization, yields REE chlorides. This method effectively mitigates HF gas during the roasting process and achieves a relatively high recovery rate for REEs. However, it has some drawbacks, including the consumption of a significant amount of acid and alkali, a complex process, higher production costs, equipment susceptibility to corrosion, poor safety performance, and the introduction of alkali metal impurities affecting the purity of the REE product. Additionally, compared to the acid method, the alkali method often requires higher ore grades [96, 124, 128–131].

It is particularly important to note that fluoride-containing wastewater must be properly treated; otherwise, it may raise concerns regarding human health, such as fluorosis, including skeletal fluorosis and dental fluorosis [132–134]. For example, during the roasting process, fluorine can be transformed into CaF_2 and Na_3AlF_6 to mitigate its environmental impact [124, 135, 136].

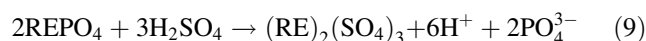
The chemical reaction equation is as follows [126]:



4.1.2.2 Monazite Monazite is a REE phosphate mineral that belongs to the monoclinic crystal system. Its chemical structure is highly stable and resistant to decomposition even at very high temperatures [137]. Furthermore, it contains a high content of REEs, ranging from 55% to 65% [93, 128, 137], making it a valuable source of REEs. Additionally, monazite is often associated with radioactive elements such as thorium, uranium, and radium [128].

Currently, there are two common methods for leaching monazite: the sulfuric acid roasting decomposition method and the alkaline leaching method [138, 139].

The essence of the sulfuric acid roasting decomposition method is to break down the stable phosphate structure of monazite through roasting with sulfuric acid [123, 140]. The key step in this method occurs within the temperature range of 200–400 °C [139], where monazite phosphate is converted into soluble REE sulfates. Subsequently, the roasted mineral is leached with water, causing REEs to precipitate as REE sulfate salts. The purification process follows, including the addition of hydroxides and carbonates to remove impurities such as iron and phosphorus from the solution while recovering REEs. Ultimately, a pure REE sulfate solution is obtained, which can be further processed by converting REE sulfate salts into REE hydroxides with sodium hydroxide and then into REE chlorides by dissolving in hydrochloric acid. This method achieves a REE recovery rate of over 97%. The sulfuric acid roasting decomposition method offers several advantages, such as simplicity, a short residence time, high REE recovery rates, and wide adaptability. However, it has drawbacks, including severe equipment corrosion, the generation of radioactive waste residues and roasting fumes, as well as challenges in extracting and recovering phosphorus from monazite [126, 137, 138]. Here is the chemical equation for the sulfuric acid roasting decomposition method [128]:

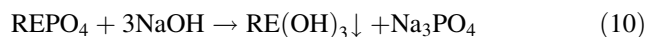


Demol et al. [140] conducted extraction research on a monazite–apatite mixture using a sulfuric acid roasting decomposition method. They found that, under a roasting temperature of 300 °C and a mass ratio of 1:1 between monazite and apatite, the leaching efficiency of REEs could reach 90%–99%.

The alkaline leaching method operates in a different route. The core of this method lies in the utilization of a high-concentration sodium hydroxide solution at elevated temperatures and pressures to convert the REE phosphorus

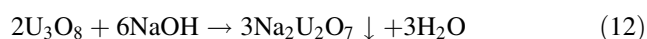
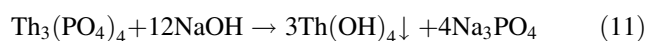
in monazite into soluble trisodium phosphate and REE hydroxides. This method can achieve a monazite recovery of up to 96%, making it economically advantageous. As a result, it has gradually replaced the sulfuric acid roasting decomposition method as the mainstream process. Notably, this technique applies to high-grade concentrates (above 60%) and does not apply to low-grade concentrates. In this method, the monazite concentrate is first finely ground and then subjected to thermal decomposition in an alkaline solution. At the same time, it usually requires high pressure (autoclave); otherwise, it dries quickly. Subsequent filtration allows for solid–liquid separation, with the liquid phase containing the trisodium phosphate mother liquor, which can be cooled and crystallized to produce trisodium phosphate products. The solid phase consists of REE hydroxides, which can be further processed through washing, acid dissolution, and concentration to obtain REE chlorides [126].

Compared to the sulfuric acid roasting decomposition method, the advantages of this method are evident. It efficiently recovers phosphorus in the form of trisodium phosphate, yielding economic benefits, and does not produce harmful gases during the production process [126]. Additionally, the alkaline leaching method allows for the recovery of thorium and uranium from monazite. However, it generates a significant amount of wastewater and consumes excess alkali. Here is the chemical equation for the alkaline leaching method [128]:



This method produces Na_3PO_4 as a marketable by-product, generating additional economic benefits [137].

In addition, the chemical reaction equations for the precipitation of thorium and uranium are as follows [128]:

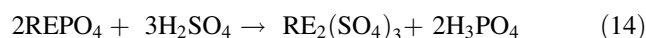
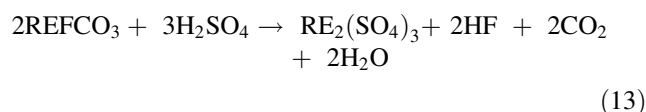


Due to the highly stable nature of monazite, both aforementioned methods require fine grinding of the mineral and the application of strong acids (e.g., 93% concentrated sulfuric acid for the sulfuric acid roasting decomposition method) or strong alkalis (e.g., 60%–75% sodium hydroxide for the alkaline leaching method). In practice, this exacerbates operating conditions and introduces risks such as equipment corrosion and unexpected accidents [137, 139].

To address these issues, Watts et al. [139] applied concentrated phosphoric acid (weak acid) to process monazite. Their study found that at 260 °C, over 99% of light REEs were dissolved within 90 min. This dissolution method avoids the utilization of strong acids or bases and reduces the risk of dust exposure caused by fine grinding.

4.1.2.3 Bayan Obo mixed rare earth ore As the host to the world's largest reserve of REE minerals, the Bayan Obo deposit in Baotou, China, contains mixed REE minerals, primarily in the form of bastnaesite and monazite [141]. The complex mineral structure of this deposit makes it challenging to process. Due to the presence of both bastnasite and monazite in the Bayan Obo REE ore, the primary methods applied to process mixed REE ore are similar to those employed in bastnasite and/or monazite individually.

Currently, the mainstream methods that are implemented in this ore type are the sulfuric acid roasting decomposition method and the alkali fusion method. Presently, over 90% of the REE concentrate from this ore is processed using the sulfuric acid roasting decomposition method. The general process involves drying the REE concentrate, mixing it with concentrated sulfuric acid, and roasting it at temperatures ranging from 700 to 800 °C to convert it into REE sulfate [142]. Subsequently, the product is leached with water to obtain a REE solution. Iron powder is added to remove phosphate ions (Fe/P mass ratio = 2–3), and then magnesium oxide is applied to adjust the pH to 3.5–4.5, resulting in a purer REE solution [115, 142]. Ammonium bicarbonate is added to precipitate a mixed REE carbonate, which is then subjected to extraction and separation, resulting in a REE chloride solution. Further solvent extraction is performed to separate different REEs. This method is characterized by its simple process, low ore concentration requirements, high applicability, low cost, and high REE recovery rates. However, due to the propensity of REE ions in sulfuric acid systems to react with monovalent alkali metal ions and ammonium ions to form insoluble REE sulfate complex precipitates, magnesium oxide powder is often added to adjust the pH during purification, introducing impurity metal ions and generating a significant amount of waste gas and wastewater. Particularly, all phosphorus (P) and thorium (Th) in the roasting process form poorly soluble salts that enter the leaching residue, posing radiation risks and requiring special attention, and the issue of phosphorus recovery. In addition, waste gases containing fluorine and sulfur are difficult to recover and utilize, and there are issues with high energy consumption and low resource utilization efficiency [141–143]. The chemical reactions are as follows:



Given the drawbacks of the sulfuric acid roasting decomposition method, Hou et al. [143]

Table 5 Technical assessment of existing mining methods for REEs

Ore type	Methods	Major reactants	Limitations	Advantages	Feasibility
Ion-adsorption REE ore	Pond leaching	–	Severe vegetation damage, large amounts of tailings, high labor intensity, low efficiency and throughput	–	Eliminated
	Heap leaching	–	High energy consumption, high labor intensity, REE recovery < 70%, radioactive residues difficult to manage	Improved productivity and reduced lixiviant consumption compared to pond leaching	Restricted, non-competitive
	Insitu leaching	–	Risk of landslides and groundwater contamination	Low labor intensity, minimal infrastructure cost, cost-effective, improved REE utilization, reduced environmental impact	Widely adopted
	–	Na ⁺ salt	High consumption, leads to soil salinization and alkalization	Low cost	Limited applicability
	–	NH ₄ ⁺ salt	Ammonia–nitrogen wastewater is difficult to treat	High leaching efficiency, low reagent consumption	Moderate
	–	Mg ²⁺ salt	Soil salinization, compaction, accumulation of Mg ²⁺ in soil	High leaching efficiency, enhanced impurity removal (13% higher than NH ₄ ⁺ method)	Preferred
	–	Ca ²⁺ + Mg ²⁺ salts	–	Matches soil nutrient ratio	Promising
	–	Multi-Mg ²⁺ salts	Soil salinization, compaction, Mg ²⁺ accumulation	High REE recovery (95.22%)	Preferred
	–	Salt-acid combination	–	Acid enhances ion exchange efficiency	Promising
	–	Bioleaching	Slow process, requires stable microbial ecosystem and nutrients	Environmentally friendly	Promising
Bastnäsite	Oxidative roasting + HCl leaching	HCl	Additional treatment is required for HF gas	Low cost, simple process, low capital investment	Widely applied
	Alkaline method	NaOH, HCl	High acid/alkali consumption, complex process, high production cost, severe equipment corrosion, safety risks, alkali metal impurities affect product purity	No HF concern, efficient REE recovery	Moderate, but favorable for ores with high alkalinity
Monazite	Sulfuric acid roasting	H ₂ SO ₄	Severe equipment corrosion, generation of radioactive residue and off-gas, difficult phosphate recovery	Simple process, short retention time, high REE recovery, adaptable	Most used
	Alkaline leaching	NaOH	Large wastewater volume, high alkali consumption, requires high temperature and pressure	Enables phosphate recovery, higher economic value, environmentally friendly, allows recovery of Th and U	Less common but promising for green applications
	Phosphoric acid method	H ₃ PO ₄	–	Avoids use of strong acids/alkalis and dust exposure from fine grinding	Promising

Table 5 continued

Ore type	Methods	Major reactants	Limitations	Advantages	Feasibility
Bayan Obo mixed REE ore (mixed with bastnaesite and monazite)	Sulfuric acid roasting	H ₂ SO ₄	Fluoride management issues, impurity introduction, air/water pollution, radiation risk	Simple process, low ore grade tolerance, broad applicability, low cost, high REE recovery	Adopted in > 90% of Bayan Obo REE operations
	Alkaline fusion	NaOH	High acid/alkali consumption, complex process, equipment corrosion, impurity impact on product purity, strict ore requirements	High REE recovery, effective removal of calcium, fluoride, and phosphate	Moderate

Table 6 REE content in some products (wt%)

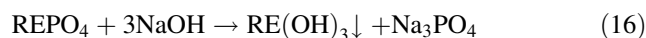
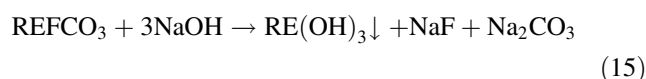
REE application	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Y	Others
Magnets	–	–	23.4	69.4	–	–	2	0.2	5	–	–
Battery alloys	50	33.4	3.3	10	3.3	–	–	–	–	–	–
Metal alloys	26	52	5.5	16.5	–	–	–	–	–	–	–
Auto catalysts	5	90	2	3	–	–	–	–	–	–	–
Petroleum refining	90	10	–	–	–	–	–	–	–	–	–
Polishing compounds	31.5	65	3.5	–	–	–	–	–	–	–	–
Glass additives	24	66	1	3	–	–	–	–	–	2	4
Phosphors	8.5	11	–	–	–	4.9	1.8	4.6	–	69.2	–
Ceramics	17	12	6	12	–	–	–	–	–	53	–
Other	19	39	4	15	2	–	1	–	–	19	–

“–” Represents absence or negligible content

developed a low-temperature sulfuric acid recycling leaching technology. This method applies a mixed REE concentrate with 75 wt% H₂SO₄ and 70 mL g^{−1} phosphoric acid for leaching and decomposition at temperatures below 200 °C. The study found that, under these conditions, both bastnaesite and monazite in the ore were completely leached.

The alkali fusion method begins by acid leaching the mineral, which removes calcium and enhances the REE concentrate's quality. Then, alkali fusion is employed to convert the ore concentrate into REE hydroxides, which are insoluble in water but soluble in acid. Simultaneously, fluorine and phosphate ions in the ore are converted into sodium fluoride and sodium phosphate, which are soluble in water, achieving an initial separation of REEs from others. After washing the REE hydroxide product with water, they are dissolved in hydrochloric acid to form a REE chloride solution. Further purification and evaporation concentration result in a REE chloride product. Alternatively, after solvent extraction, different REEs can be

separated, yielding individual REE products [115, 142]. The chemical reaction equations are as follows:



4.1.3 Summary

In this section, a systematic overview of mainstream extraction strategies for various types of REE ores is presented. Each approach is critically evaluated with respect to the commonly employed chemical reagents, operational advantages, inherent limitations, and overall technical feasibility. A comparative summary is provided in tabular form (Table 5) to enhance the clarity and accessibility of current industrially relevant methodologies, focusing on widely adopted and representative techniques.

After leaching, the resulting metal-containing solution can be further processed to achieve the separation of

individual elements based on differences in their chemical properties. Common separation techniques include adsorption, precipitation, and solvent extraction methods [144, 145].

Commonly available solvent extractants can be classified into acidic and neutral types. Acidic extractants mainly include organic acids and organophosphoric acids, such as naphthenic acid (NA), di(2-ethylhexyl) phosphoric acid (D2EHPA), 2-ethylhexylphosphonic acid mono-2-ethylhexyl ester (PC88-A), di(2,4,4-trimethylpentyl) phosphoric acid (Cyanex 272), bis(2,4,4-trimethylpentyl)phosphinic acid (Cyanex 572), Reinecke acid, and 2-(bis(2-ethylhexyl)oxy)phosphoryl-2-hydroxyacetic acid (HPOAc) [145, 146].

4.2 Recycling

As noted previously, in addition to primary production routes of REEs through mining, recycling is also a way to obtain REEs. Recycling can solve the problem of an unstable REE supply chain and alleviate the fluctuation of REE market prices [17]. Especially for HREEs, not only are the reserves getting smaller and the gap getting bigger, but they also face a non-renewable situation. As a result, recycling is inevitable [33].

Currently, the goals of recycling can be roughly divided into three targets: (1) REE tailings; (2) waste REE products; and (3) others [36].

Tailings are a highly valuable and underutilized rare earth resource. Due to the usage of underdeveloped technology for extracting REEs in the previous periods, the tailings of REE deposits contain high concentrations of REEs. For example, the REO content of the tailings dam at Bayan Obo is about 3%–5%, and large amounts of REEs are also contained in the tailings of some other minerals. In summary, the amount of REEs in tailings areas cannot be ignored [36], so it is crucial, inevitable, and prospective to recover REEs from tailings.

However, the REE grade in tailings is low, with a large amount of gangue, and valuable metals often exist in fine particle forms within the ore, posing challenges for sensing, separation, and extraction processes. Furthermore, tailings frequently contain a high proportion of fine particles smaller than 150 μm , which are difficult to process and lack mature concentration methods [11, 147]. These factors create challenges for tailings recycling. While some methods, such as hydrometallurgical techniques, are effective, they are far from being ready for industrial application [11]. The widespread adoption of tailings recovery depends on its ability to generate economic benefits, which will undoubtedly drive the development of the industry. In fact, mineral separation and concentration are key; if these issues can be addressed and tailings can be enriched to a grade similar to that of the primary ore using simple, cost-effective, and pollution-free methods, the same metallurgical processes used for the initial ore can be applied to recover the valuable minerals.

There are many end products which constitute REEs as potential recyclable fractions. In particular, the REE content in some products is shown in Table 6. If the REEs in these materials can be efficiently recovered, it can turn waste into wealth and bring substantial resources and economic benefits.

As another major source of recyclable REEs, e-waste is the fastest-growing type of end product at an increasing rate of 20–50 million tons per year, accounting for about 5% of urban waste. Of all scrapped products containing REEs, e-waste has received the most attention in recent years worldwide [148]. Recycling REEs from e-waste is considered a sustainable alternative for the REE industry, helping to compensate for the limited availability of natural REE resources and the challenges associated with their extraction [149]. However, currently, less than 1% of REEs are recycled and utilized [150].

For e-waste, the initial step involves pretreatment, including processes such as disassembly, crushing, grinding, and physical separation. In pyrometallurgical

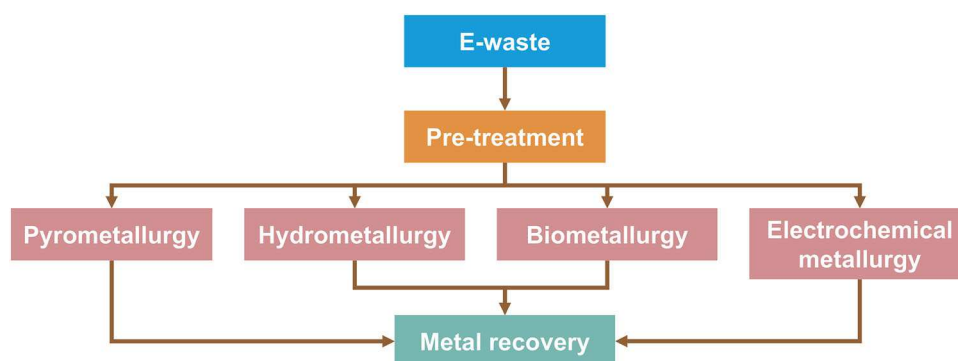


Fig. 10 Flowchart of REEs recycling from e-waste

processes, relatively coarse particles are typically sufficient, while for hydrometallurgical processes, finer grinding to a size of less than 200 μm is often required [149, 151, 152].

The next step is the physical separation process, which is essentially the beneficiation process. Common techniques include vibration screening, density-based separation, conductivity-based separation, magnetic susceptibility, optical sorting, and separation based on brittleness and phase hydrophobicity. Specifically, magnetic separation can be utilized to remove ferrous metals, followed by the recovery of non-magnetic metals through pyrometallurgical, hydrometallurgical, biometallurgical, or electrometallurgical methods. Among these, pyrometallurgy and hydrometallurgy are the most adopted methods [149, 153–155]. The relevant e-waste recycling flow diagram is shown in Fig. 10.

When the e-waste obtained after beneficiation is considered a special REE mineral. Therefore, the recycling process of REEs and the metallurgy process of REE ores can be viewed as fundamentally similar. Thus, the metallurgy process for REE recycling from e-waste follows the main steps outlined in the previous chapter on mining processes. These steps include roasting (pyrometallurgical), leaching, impurity removal, separation, and precipitation of REEs. Since the chemical properties of REEs remain unchanged in initial ores, tailings, or e-waste, the techniques discussed in the previous chapter can be referred to as a guideline.



Fig. 11 Tai Chi sustainability model of the rare earth industry demand

In addition to tailings and e-waste, industrial by-products, metallurgical residue, wastewater, and other materials are also potential sources for REE recycling and cannot be overlooked.

Recovering REEs from these three major sources helps ease the burden on limited REE resources, alleviates supply–demand conflicts, and reduces the natural disasters and environmental pollution caused by mining and metallurgical processes, such as pit collapses, landslides, as well as air and water body contamination. This benefits the sustainable development of the REE industry. Additionally, recycling of e-waste and tailings, on the one hand, reduces environmental pollution caused by their leakage, and on the other hand, prevents them from occupying land resources due to accumulation [149, 156, 157]. Overall, the recycling process transforms waste into wealth, which is vital to the sustainable development of REEs. However, considering that the overall recycling rate of REEs is below 1%, the REE recycling industry still faces significant challenges.

REE recycling accounted for less than 5% of global REE supply [36]. With the extensive application of REEs in the future, there will inevitably be a large amount of e-waste and retired REE products. Therefore, the recycling of secondary resources is full of development prospects [48].

In addition to traditional metallurgical recovery methods, REE-containing e-waste and tailings can be directly processed into corresponding products, such as using REE tailings to make pigments [157]. Furthermore, by using plants to restore mining areas, not only can the ecological environment be restored, but valuable metals can also be recovered from low-grade ores (tailings) through phytoremediation. Although REEs are not essential for plants, they can accumulate in plant tissues at moderate concentrations, a process known as phytomining [158]. These non-metallurgical methods also contribute to REE recovery, as they not only avoid environmental pollution and ecological imbalance but also turn waste into wealth, alleviate resource scarcity, bring economic benefits, and promote the sustainable development of the REE industry.

5 Key factors, challenges, and prospects

5.1 Key factors for sustainability

In the context of increasing ecological fragility, the widespread adoption of environmental principles, the depletion of high-quality mineral resources, and a volatile international landscape affecting global resource industries and supply–demand chains, sustainability can be seen as a key development goal across many industries, including REEs.

Given REEs' strategic importance, they are crucial for scientific and technological advancements as well as societal transformations. However, the sustainability of the REE industry is influenced by multiple factors, each playing distinct roles and responsibilities within REE development. These factors impact and constrain each other, and the absence of any one of them could trigger chain reactions with significant effects on the entire sector.

To better visualize the importance, relationships, and mutual influences among these different components, we introduce a Tai Chi model that leverages the Tai Chi (Yin-Yang) philosophy to represent these interconnected elements (Fig. 11). Our Tai Chi model includes one central goal (representing the Tai Chi itself), two pathways (the two aspects of Yin and Yang), four core entities (the four elements), and eight influencing factors (the eight trigrams). Below, we will elaborate on these elements individually.

(1) Tai Chi: one goal

At the center of the model, the sustainable development of the REE industry is the core objective. This requires consideration of sustainable methods for managing and developing REE resources to ensure their long-term viability, mitigating the environmental and economic impacts of over-extraction and excessive usage. Achieving sustainability necessitates a comprehensive industry framework. Our Tai Chi model includes two methods, four entities, and eight factors, all of which are interconnected and mutually influential in driving the sustainable progress of the REE industry. Sustainability represents a collective aspiration, symbolizing a balanced development across both approaches, harmonious interconnection among the four core entities, and equal consideration for the eight factors, ensuring a balanced, resilient REE sector.

(2) Two aspects: two pathways

These two pathways can be represented by the "Yin-Yang" concept within Tai Chi, symbolizing "mining" and "recycling", the two primary methods for acquiring REEs. Mining is focused on primary resources, i.e., fresh mineral deposits, whereas recycling emphasizes secondary resources and tailings, serving as an effective supplement to mining. Despite their different sources, these two methods are complementary, forming the foundation of REE supply. Mining, as the common approach to REE resource development, must balance environmental and societal costs, while recycling waste and tailings, helping to better manage REE resources and achieve sustainability. Although mining targets primary REE resources and recycling addresses secondary REE resources, both are inseparable, much like two legs supporting balance. Together, they underpin REE supply, sustainability, and the development of the sector, reflecting the dual nature of Tai Chi.

(3) Four elements: four social roles

In our Tai Chi model, the four elements represent four social roles that influence REE development: government, enterprises, researchers, and consumers. Each of these actors plays a distinct role and bears unique social responsibilities in the sustainable development of REEs.

(a) Governments: The government's responsibilities encompass both domestic and international domains. Domestically, it needs to create policies and regulations to oversee REE extraction and utilization, and implement incentives or penalties to encourage companies to engage in low-cost production and address technological challenges. Through tax incentives and funding support, governments can also promote the domestic market order, combat illegal mining, phase out outdated capacity, and foster environmental education to raise public awareness about recycling. Internationally, the government plays a role in global REE sustainability through cooperation, resource supply management, trade regulation, and sharing or protecting advanced technologies to enhance national competitiveness and influence. For developing countries with limited REE reserves or technological capacity, international partnerships can be particularly challenging yet essential. Intergovernmental collaboration and competition shape the global REE market's sustainability, while technology and resource sharing support the broader development of the industry.

(b) Enterprises: Companies play a crucial role in transforming REE resources into products and in recycling REE waste. From primary resource extraction to final products, efficient, safe operations that produce market-ready REE products are essential for a stable industry. In recycling, companies must promptly and efficiently recover discarded REE products to prevent resource waste and environmental harm. For sustainable development, enterprises should consider energy consumption, environmental impact, economic feasibility, and social responsibility, striving for efficient and clean production. Transparency is also key, because reliable products with accurate specifications could build consumer trust and support ongoing demand. Collaboration among companies, including joint ventures, technology sharing, and avoiding destructive competition, is beneficial to the sector's advancement, as sharing patents, databases, and platforms can accelerate the REE industry's progression.

(c) Researchers: Researchers drive technological innovation, introducing competitive new methods and refining existing ones to enhance the efficiency of primary and secondary REE resource utilization, reduce environmental pollution, cut costs, and explore REE alternatives. Researchers are key to the REE industry's sustainable breakthroughs and technological evolution.

(d) Consumers: Consumer choices and demands directly affect REE supply and demand. As the end-users,

consumers' responsible usage and recycling of REE-related products are also crucial to sustainability.

Although each actor fulfills different roles with specific duties, sustainable development in the REE industry relies heavily on robust collaboration and dialog among these four participants. The absence or separation of anyone would inflict major, potentially devastating, disruptions on the sector.

For instance, government regulations influence corporate production practices and can guide the direction of new technological developments. Stricter environmental policies often drive cleaner production technologies, thereby impacting researchers. Government policies and funding also play a critical role in enabling researchers to develop new technologies. Enterprises, in turn, depend on researchers' findings to maintain competitive advantage, and partnerships between research institutions and companies are increasingly common. Transitioning technology from laboratory research to industrialization requires collaboration between researchers and enterprises, including production experience, capital, facilities, and equipment that support scale-up efforts.

A stable REE market requires a supply–demand balance between enterprises and consumers, with appropriate government oversight. Consumer behavior drives corporate production, and companies adjust their production to meet consumer demands, thus preventing surpluses or shortages. Consumer feedback to the government is also valuable, enabling timely policy adjustments. Recycling of secondary resources depends on consumers collecting and returning discarded REE products, which companies can then reclaim.

Table 7 Evolution of the resource tax on rare earths in China [159]

Period	Resource tax
Before 1994	The tax rate was 0, REE mining for free of charge
1994.01–2011.03	Tax rate by volume was 0.4–30 CNY per ton according to non-ferrous metal minerals, actual tax rate was 5.66 CNY per ton for light REE raw ore while 2.00 CNY per ton for heavy REE raw ore
2011.04–2015.04	The tax rate by volume was 60 CNY per ton for light REE raw ore and 30 CNY per ton for heavy REE raw ore
After 2015.05	Tax rate by ad valorem is 11.50%, 9.50%, and 7.50% for light REE concentrate ore in Inner Mongolia, Sichuan and Shandong while 27.00% for heavy REE concentrate ore

These interactions underscore the indispensable roles of all four actors in achieving REE industry sustainability.

(4) Eight Trigrams: eight influential factors

In the Tai Chi model, the eight trigrams symbolize the eight factors influencing the development of REEs: resource, energy, environment, policy, application, technology, supply and demand, and economy. These factors play a crucial role in sustainability.

Resources form the foundation for the extraction and utilization of REEs. This includes both primary and secondary resources, corresponding to the two pathways of mining and recycling. Additionally, the uneven global distribution of REE resources affects their sustainability, with some countries potentially facing resource scarcity. Different resource types require different technological approaches and exhibit distinct economic attributes and production costs. These two points highlight the close interrelation between resources, technology, and the economy. Furthermore, the abundance and extraction difficulty of the resource directly influence supply & demand dynamics.

Energy plays a dual role in the REE industry. On the one hand, the mining and recycling processes of REEs are accompanied by energy consumption, making energy efficiency a central concern for sustainability. On the other hand, REE materials also find applications in the energy sector, as shown in Table 2 [17, 36, 46, 47]. Thus, energy is both an input and output factor in the lifecycle of REEs. Balancing energy consumption while supporting clean energy applications is crucial for REE sustainability.

The continuous advancement of technology is essential to enhancing the sustainable supply of REEs. This helps address more complex and stringent environmental concepts and policies, more complicated minerals, and the increasing accumulation of retired e-waste containing REEs. Currently, less than 1% of REEs are recycled [17, 150], which indicates that technological innovation is necessary not only in the mining sector but also in the recycling sector to achieve sustainable resource circulation. Moreover, technological progress can lead to lower production costs, higher efficiency, better product performance, more intelligent production processes, and environmentally friendly production techniques. Furthermore, technology impacts both the environment and resources. Outdated technologies may lead to environmental burdens and inefficient or wasteful resource utilization, while sustainable technologies typically exhibit environmental benefits and can process low-grade or difficult-to-extract REE resources. Conversely, environmental degradation and resource depletion drive technological innovation.

Policy directly influences the production methods, output, and pricing of REEs, with its impact being critical.

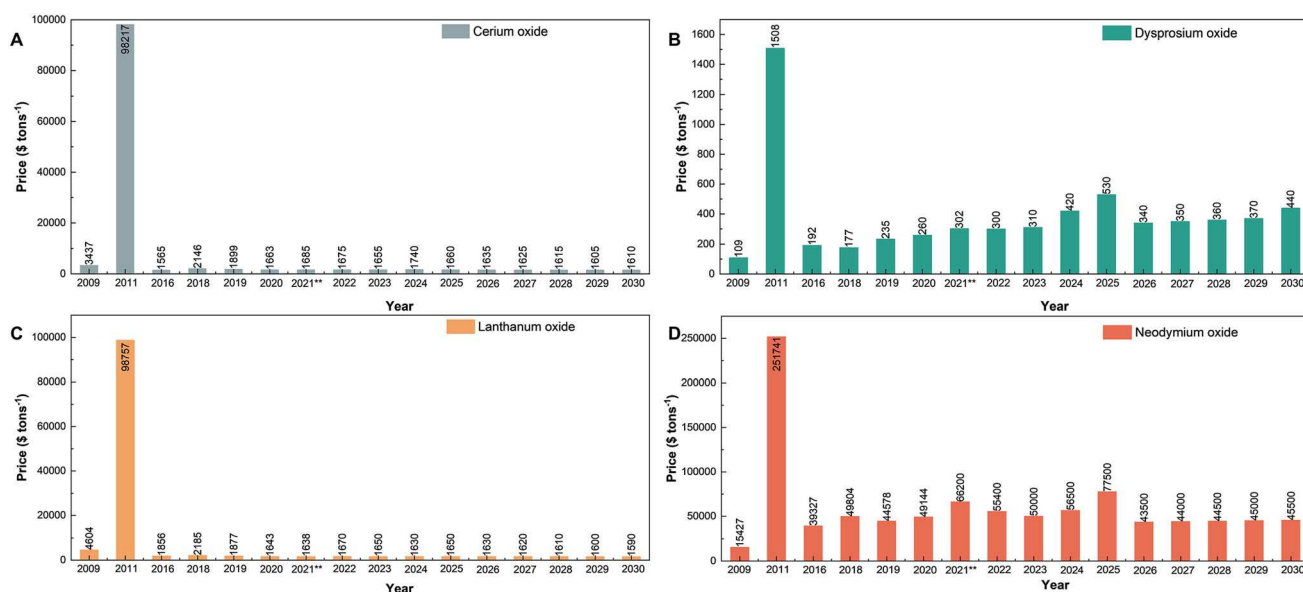


Fig. 12 Several rare earth oxide price changes: **a** cerium oxide; **b** dysprosium oxide; **c** lanthanum oxide; **d** neodymium oxide (data from Ref [166])

Policies not only affect specific production technologies but also influence the global REE market. For example, increasingly stringent environmental policies gradually phase out outdated capacities and environmentally hazardous reagents, crack down on illegal activities, impose resource taxes, and promote industrial consolidation, which fosters the development of the REE industry and prevents resource misuse, as reflected in China's REE sector [12, 159–164]. The evolution of China's REE resource tax is shown in Table 7. Additionally, China's production quotas led to a significant increase in REE prices in 2011, as illustrated in Fig. 12 [12, 165, 166]. Policy decisions are typically based on in-depth analysis and assessment of factors such as resources, energy, environment, supply and demand, technology, and economy. In turn, policies influence resource extraction, energy consumption, environmental standards, market balance, technological development, and economic conditions.

Application is the ultimate driving force behind the REE industry; without practical applications, the REE industry would not exist. The wider application of REEs, coupled with the emergence of alternative materials, represents a sustainable trend. Applications drive production, promote mining processes, influence supply and demand, and the economic aspects. The use of REEs in high-tech industries also demonstrates the interrelationship between application and technology.

Supply & demand and economic conditions affect the market stability and pricing of REEs, ultimately influencing the production volume of REEs. Only a balanced supply and demand relationship and a healthy REE economy can ensure the sustainability of the industry.

Moreover, supply and demand and economic factors also influence policymaking.

In summary, these eight factors identified each play an indispensable role. More importantly, they are deeply interconnected, mutually reinforcing and constraining one another. Together, they form a comprehensive framework for assessing and guiding the sustainable development of the REE industry.

5.2 Challenges

Currently, although REEs have been widely applied and their usage continues to grow year by year, their development still faces many challenges:

- (1) The concentration of REEs is not uniform and differs greatly, with differences in abundance reaching several hundred times. For example, in the earth's crust, Ce is 63 ppm and La is 31 ppm, while Tm and Lu have contents of only 0.3 ppm and 0.31 ppm, respectively [16]. Pm is almost non-existent in the Earth's crust [19]. Therefore, some REEs are actually scarce, and there is still a supply risk in the long term.
- (2) Due to the extremely dispersed distribution of REEs and their low concentration for mining, it is difficult to extract them efficiently and economically using environmentally friendly methods [49].
- (3) REEs have similar chemical properties, making it difficult to separate them from each other [34, 53].
- (4) REE mining can also produce pollutants such as dust and wastewater, causing a range of environmental problems. Increasingly strict environmental

policies and high environmental management costs have also become obstacles to stable REE supply [17, 167].

- (5) Poorly managed REE mining operations may also cause geological disasters such as mine collapses and landslides [34, 46, 50].
- (6) REE minerals often coexist with radioactive elements such as uranium and thorium, which can threaten human life and safety [17, 35].
- (7) In some cases, recovery rates of REEs are very low, which can be less than 1%, causing a lot of resource waste. The development of efficient methods of REE recovery still has a long way to go [17].
- (8) REE processing is dominated by a few countries, and changes in the policies of major REE reserve-producing countries can also cause fluctuations in the global REE market supply [80, 168].
- (9) Treatment of REE mine residues: for example, processing one ton of Bayan Obo REE concentrate generates approximately 1.4–1.6 tons of residues. The treatment of these residue requires additional financial investment [141].
- (10) In some countries and regions, inadequate laws and regulations have led to illegal mining of REE resources, resulting in significant resource waste and associated environmental issues [169–171].
- (11) The lack of synchronization in global technology has introduced uncertainty in the supply, demand, and market of REEs [16, 70, 172].

5.3 Prospects

Facing these challenges, we recommend taking measures to guide the healthy, benign, and sustainable development of the REE industry. These measures include, but are not limited to, the following:

- (1) Improve laws and regulations, establish strong incentives such as encouraging investment in the REE field and promoting technological innovation, and punishment measures, cracking down on illegal mining activities [12, 173].
- (2) Strengthen research and development of novel technologies and relevant equipment. Generally, these novel technologies should have the following characteristics: development/utilization of cleaner, environmentally friendly reagents, faster routes in production, higher efficiency, and less environmental pollution [12, 174–176].
- (3) Promote international cooperation, reduce disputes, and form complementary advantages [1, 175, 177].
- (4) Develop REE alternative materials to reduce dependence on REEs [1].

- (5) Extend the service life of REE products.
- (6) Improve the REE recovery rates from tailings, e-waste, and other secondary resources [12].
- (7) Safeguard REE reserves, establish more diversified supply channels, and make contingency plans and remedial measures to deal with emergencies in the process of production and application [1, 175, 176].
- (8) Extend and strengthen the REE industry chain and vigorously develop upstream and downstream industries [1, 174, 177].

6 Conclusion

This paper provides a systematic analysis of the history, current status, and future prospects of rare earth elements (REEs), focusing on key issues related to their chemistry, applications, resources, and processing technologies. Through an analysis of global REE resource distribution and an in-depth examination of China's REE industry, the paper provides a comprehensive overview of the current global market of REEs and highlights both technological and non-technological opportunities and challenges facing the sector. The paper reviews major REE-bearing minerals such as ion-adsorption type REE ores, bastnaesite, monazite, and Bayan Obo mixed rare earth ore, detailing the extraction technologies tailored to each type of ore and summarizing recent advancements in the field. In addressing sustainable development, the paper introduces a philosophical “Tai Chi Model” which conceptualizes two complementary production pathways—mining and recycling—and explores the collaborative roles of four key stakeholders: governments, enterprises, researchers, and consumers. Additionally, the paper identifies eight critical and interconnected factors—resources, energy, environment, policy, applications, technology, supply and demand, and economics—that shape the future of the REE industry. By analyzing these current challenges, this paper offers concrete ideas and practical strategies to promote sustainable development. These include strengthening legal and regulatory systems, enhancing technological innovation, fostering international collaboration, and developing substitute materials. Overall, this review not only provides a comprehensive understanding of the REE industry but also offers theoretical guidance for policymaking and technological innovation, providing valuable insights for advancing sustainable development in this strategically important field.

Acknowledgements The authors appreciate all the researchers whose articles were used in the present research. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.



Author contributions T-Y Zhao was responsible for investigation, writing original draft, conceptualization, writing—review and editing, and project administration. W-L Li took part in investigation, conceptualization, writing—review and editing, and project administration. S Kelebek was involved in investigation, conceptualization, and writing—review and editing. Y Choi, C-Q Wu, W-J Zhang, and C-Y Wang performed writing—review and editing. Z-W Zhao was responsible for funding acquisition, project administration, and writing—review and editing. F Sadri participated in project administration and writing—review and editing.

Data availability statement The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interests The authors declare that they have no conflict of interest.

References

- [1] Hu X, Sun B, Wang C, Lim MK, Wang P, Geng X, Yao C, Chen W. Impacts of China's exports decline in rare earth primary materials from a trade network-based perspective. *Resour Policy*. 2023;81:103321. <https://doi.org/10.1016/j.resourpol.2023.103321>.
- [2] Zhou H, Wang J, Yu X, Kang J, Qiu G, Zhao H, Shen L. Effective extraction of rare earth elements from ion-adsorption type rare earth ore by three bioleaching methods. *Sep Purif Technol*. 2024;330:125305. <https://doi.org/10.1016/j.seppur.2023.125305>.
- [3] Zhu M, Sun X, Liu R, Xu H. Development strategies for rare earth functional materials by 2035. *Strategic Study CAE*. 2020; 22(5):37. <https://doi.org/10.15302/J-SSCAE-2020.05.006>.
- [4] Pereto C, Baudrimont M, Coyne A. Global natural concentrations of rare earth elements in aquatic organisms: progress and lessons from fifty years of studies. *Sci Total Environ*. 2024; 922:171241. <https://doi.org/10.1016/j.scitotenv.2024.171241>.
- [5] Pang X, Li DC, Peng A. Application of rare-earth elements in the agriculture of China and its environmental behavior in soil. *Environ Sci Pollut Res*. 2002;9(2):143. <https://doi.org/10.1007/bf02987462>.
- [6] Schulz KJ, Deyoung JH, Seal RR, Bradley DC. Critical mineral resources of the United States—economic and environmental geology and prospects for future supply. Professional paper. 2017;1802. <https://doi.org/10.3133/pp1802>.
- [7] Dutta T, Kim KH, Uchimiya M, Kwon EE, Jeon BH, Deep A, Yun ST. Global demand for rare earth resources and strategies for green mining. *Environ Res*. 2016;150:182. <https://doi.org/10.1016/j.envres.2016.05.052>.
- [8] An J, Qu Y, Wang G. The multiple roles of rare earth elements in the field of photocatalysis. *Inorganic Chem Front*. 2023; 11(1):11. <https://doi.org/10.1039/d3qi02006a>.
- [9] Humphries M. Rare earth elements: The global supply chain (CRS). In: Federation of American Scientists Congressional research Service. 2013;1–31. <https://sgp.fas.org/crs/natsec/R41347.pdf>.
- [10] Jiang Y, Fu H, Liang Z, Zhang Q, Du Y. Rare earth oxide based electrocatalysts: synthesis, properties and applications. *Chem Soc Rev*. 2024;53(2):714. <https://doi.org/10.1039/d3cs00708a>.
- [11] Abbadi A, Mucsi G. A review on complex utilization of mine tailings: recovery of rare earth elements and residue valorization. *J Environ Chem Eng*. 2024;12(3):113118. <https://doi.org/10.1016/j.jece.2024.113118>.
- [12] Mancheri NA, Sprecher B, Bailey G, Ge J, Tukker A. Effect of Chinese policies on rare earth supply chain resilience. *Resour Conserv Recycl*. 2019;142:101. <https://doi.org/10.1016/j.resconrec.2018.11.017>.
- [13] Goodenough KM, Wall F, Merriman D. The rare earth elements: demand, global resources, and challenges for resourcing future generations. *Nat Resour Res*. 2018;27(2):201. <https://doi.org/10.1007/s11053-017-9336-5>.
- [14] Arshi PS, Vahidi E, Zhao F. Behind the scenes of clean energy: the environmental footprint of rare earth products. *ACS Sustain Chem Eng*. 2018;6(3):3311. <https://doi.org/10.1021/acsschemeng.7b03484>.
- [15] Jyothi RK, Thenepalli T, Ahn JW, Parhi PK, Chung KW, Lee JY. Review of rare earth elements recovery from secondary resources for clean energy technologies: grand opportunities to create wealth from waste. *J Clean Prod*. 2020;267:122048. <https://doi.org/10.1016/j.jclepro.2020.122048>.
- [16] Weng Z, Jowitt SM, Mudd GM, Haque N. A detailed assessment of global rare earth element resources: opportunities and challenges. *Econ Geol*. 2015;110(8):1925. <https://doi.org/10.2113/econgeo.110.8.1925>.
- [17] Zaimes GG, Hubler BJ, Wang S, Khanna V. Environmental life cycle perspective on rare earth oxide production. *ACS Sustain Chem Eng*. 2015;3(2):237. <https://doi.org/10.1021/sc500573b>.
- [18] Golroudbary S, Makarava I, Kraslawski A, Repo E. Global environmental cost of using rare earth elements in green energy technologies. *Sci Total Environ*. 2022;832:155022. <https://doi.org/10.1016/j.scitotenv.2022.155022>.
- [19] Zhou B, Li Z, Chen C. Global potential of rare earth resources and rare earth demand from clean technologies. *Minerals*. 2017;7(11):203. <https://doi.org/10.3390/min7110203>.
- [20] Li XY, Ge JP, Chen WQ, Wang P. Scenarios of rare earth elements demand driven by automotive electrification in China: 2018–2030. *Resour Conserv Recycl*. 2019;145:322. <https://doi.org/10.1016/j.resconrec.2019.02.003>.
- [21] Guo Q, Wu Y. Assessing China's rare earth supply security. *Resour Policy*. 2024;95:105177. <https://doi.org/10.1016/j.resourpol.2024.105177>.
- [22] Chen W, Wang P, Meng FR, Pehlken A, Wang QC, Chen WQ. Reshaping heavy rare earth supply chains amidst China's stringent environmental regulations. *Fundam Res*. 2024. <https://doi.org/10.1016/j.fmre.2023.11.019>.
- [23] Vekasi K. Politics, markets, and rare commodities: responses to Chinese rare earth policy. *Jpn J Polit Sci*. 2019;20(1):2. <https://doi.org/10.1017/s1468109918000385>.
- [24] Lee Y, Dacass T. Reducing the United States' risks of dependency on China in the rare earth market. *Resour Policy*. 2022;77: 102702. <https://doi.org/10.1016/j.resourpol.2022.102702>.
- [25] Lai D, Fang W, Wu Y, Tan J. The impact of US, Japan and Europe's strategy of reconstructing rare earth supply chain on China's counterpart. *Sci Technol Rev*. 2022;40(21):88. <https://doi.org/10.3981/j.issn.1000-7857.2022.21.009>.
- [26] Wang X, Zhou J, Chi Q, Wang W, Zhang B, Nie L, Liu D, Xu S, Wu H, Gao Y. Geochemical background and distribution of rare earth elements in China: implications for potential prospects. *Acta Geosci Sin*. 2020;41(6):747. <https://doi.org/10.3975/cagsb.2020.102802>.
- [27] Zhao X, Guillen TR, Graham TR, Chen P, Zhao Y, Mergelsberg ST, Zhu Z, Wang Z, Rosso KM, Guo X, Zhang X. Incorporation of rare earth elements Sc, Y and La into gibbsite. *Chem Eng J*. 2024;485:149809. <https://doi.org/10.1016/j.cej.2024.149809>.
- [28] Xue HY, Lv GZ, Wang L, Zhang TA. Review of rare earth extraction and product preparation technologies and new

- thinking for clean utilization. *Miner Eng.* 2024;215:108796. <https://doi.org/10.1016/j.mineng.2024.108796>.
- [29] Greinacher E. History of rare earth applications, rare earth market today: Overview. *ACS Symp Ser.* 1981;1(1):3.
- [30] Reddy RR. Prospective potential applications and emerging tendencies in rare earth materials. *Mater Res Found.* 2024;164:177. <https://doi.org/10.21741/9781644903056-4>.
- [31] Behrsing T, Deacon GB, Junk P. The chemistry of rare-earth metals. In: Liddle ST, Mills DP, Natrajan LS, editors. *Lanthanides and actinides, the: synthesis, reactivity, properties and applications*. Munich: World Scientific Publishing Europe Ltd; 2021. 1–36. https://doi.org/10.1142/9781800610163_0001.
- [32] Wang J, Guo M, Liu M, Wei X. Long-term outlook for global rare earth production. *Resour Policy.* 2020;65:101569. <https://doi.org/10.1016/j.resourpol.2019.101569>.
- [33] Liu T, Chen J. Extraction and separation of heavy rare earth elements: a review. *Sep Purif Technol.* 2021;276:119263. <https://doi.org/10.1016/j.seppur.2021.119263>.
- [34] Stegen KS. Heavy rare earths, permanent magnets, and renewable energies: an imminent crisis. *Energy Policy.* 2015;79:1. <https://doi.org/10.1016/j.enpol.2014.12.015>.
- [35] Klingner JM. A historical geography of rare earth elements: from discovery to the atomic age. *Extr Ind Soc.* 2015;2(3):572. <https://doi.org/10.1016/j.exis.2015.05.006>.
- [36] Dushyantha N, Batapola N, Ilankoon IMSK, Rohitha S, Premasiri R, Abeysinghe B, Ratnayake N, Dissanayake K. The story of rare earth elements (REEs): occurrences, global distribution, genesis, geology, mineralogy and global production. *Ore Geol Rev.* 2020;122:103521. <https://doi.org/10.1016/j.oregeorev.2020.103521>.
- [37] Nassar NT, Du X, Graedel TE. Criticality of the rare earth elements. *J Ind Ecol.* 2015;19(6):1044. <https://doi.org/10.1111/jiec.12237>.
- [38] Tao Y, Shen L, Feng C, Yang R, Qu J, Ju H, Zhang Y. Distribution of rare earth elements (REEs) and their roles in plant growth: a review. *Environ Pollut.* 2022;298:118540. <https://doi.org/10.1016/j.envpol.2021.118540>.
- [39] Elkina V, Kurushkin M. Promethium: to strive, to seek, to find and not to yield. *Front Chem.* 2020;8:588. <https://doi.org/10.3389/fchem.2020.00588>.
- [40] Khan AM, Abu Bakar NK, Abu Bakar AF, Ashraf MA. Chemical speciation and bioavailability of rare earth elements (REEs) in the ecosystem: a review. *Environ Sci Pollut Res.* 2017;24(29):22764. <https://doi.org/10.1007/s11356-016-7427-1>.
- [41] Sager M, Wiche O. Rare earth elements (REE): origins, dispersion, and environmental implications—a comprehensive review. *Environments.* 2024;11(2):24. <https://doi.org/10.3390/environments11020024>.
- [42] Dai T, Gao T, Wen B. China's rare earth supply and demand pattern and balanced utilization strategy from perspective of elements. *Bull Chin Acad Sci.* 2022;37(11):1586. <https://doi.org/10.16418/j.issn.1000-3045.20220830002>.
- [43] Xu T, Zheng X, Ji B, Xu Z, Bao S, Zhang X, Li G, Mei J, Li Z. Green recovery of rare earth elements under sustainability and low carbon: a review of current challenges and opportunities. *Sep Purif Technol.* 2024;330:125501. <https://doi.org/10.1016/j.seppur.2023.125501>.
- [44] Virginia Department of Energy. Rare Earth Elements. 2024; <https://energy.virginia.gov/geology/REE.shtml>. Accessed 18 May 2025.
- [45] Hossain MK, Khan MI, El-Denglawey A. A review on biomedical applications, prospects, and challenges of rare earth oxides. *Appl Mater Today.* 2021;24:101104. <https://doi.org/10.1016/j.apmt.2021.101104>.
- [46] Ganguli R, Cook DR. Rare earths: a review of the landscape. *MRS Energy Sustainab.* 2018;5:E9. <https://doi.org/10.1557/mre.2018.7>.
- [47] Tian L, Chang H, Tang P, Li T, Zhang X, Liu S, He Q, Wang T, Yang J, Bai Y, Vidic RD, Crittenden JC, Liu B. Rare earth elements occurrence and economical recovery strategy from shale gas wastewater in the Sichuan Basin, China. *ACS Sustain Chem Eng.* 2020;8(32):11914. <https://doi.org/10.1021/acssuschemeng.0c04971>.
- [48] Alves Dias P, Bobba S, Carrara S, Plazzotta B. The role of rare earth elements in wind energy and electric mobility—an analysis of future supply/demand balances. 2020;1–33. <https://doi.org/10.2760/303258>.
- [49] Ferreira G, Critelli J. China's global monopoly on rare-earth elements. *US Army War Coll Q Param.* 2022;52(1):57. <https://doi.org/10.55540/0031-1723.3129>.
- [50] Huang X, Zhang G, Pan A, Chen F, Zheng C. Protecting the environment and public health from rare earth mining. *Earths Future.* 2016;4(11):532. <https://doi.org/10.1002/2016ef000424>.
- [51] Yan G, Li Z, Zhou B. Global political economy of rare earths: changing positions of major market actors including China, European Union, Japan and United States. In: *Proceedings of the 5th international conference on energy materials and environment engineering (ICEMEE)*, Kuala Lumpur, MALAYSIA, 2019; 052022. <https://doi.org/10.1088/1755-1315/295/5/052022>.
- [52] Kiggins RD, editor. *The strategic and security implications of rare earths*. London: Palgrave Macmillan; 2015.1–19. https://doi.org/10.1057/9781137364241_1.
- [53] Klingelhoefer D, Braun M, Droege J, Fischer A, Brueggmann D, Groneberg DA. Environmental and health-related research on application and production of rare earth elements under scrutiny. *Glob Health.* 2022;18(1):86. <https://doi.org/10.1186/s12992-022-00879-5>.
- [54] Zhang A, Ge J. Rare earth trade status and future trend between China and US. *Sci Technol Rev.* 2022;40(21):77. <https://doi.org/10.3981/j.issn.1000-7857.2022.21.008>.
- [55] Paulick H, Machacek E. The global rare earth element exploration boom: an analysis of resources outside of China and discussion of development perspectives. *Resour Policy.* 2017;52:134. <https://doi.org/10.1016/j.resourpol.2017.02.002>.
- [56] Wang P, Yang YY, Heidrich O, Chen LY, Chen LH, Fishman T, Chen WQ. Regional rare-earth element supply and demand balanced with circular economy strategies. *Nat Geosci.* 2024;17(1):94. <https://doi.org/10.1038/s41561-023-01350-9>.
- [57] Daniel JC. Rare Earths. U.S. Geological Survey, Mineral Commodity Summaries. January 2024. <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-rare-earths.pdf>. Accessed 18 May 2025.
- [58] Balaram V. Rare earth elements: a review of applications, occurrence, exploration, analysis, recycling, and environmental impact. *Geosci Front.* 2019;10(4):1285. <https://doi.org/10.1016/j.gsf.2018.12.005>.
- [59] Chen P, Ilton ES, Wang Z, Rosso KM, Zhang X. Global rare earth element resources: a concise review. *Appl Geochem.* 2024;175:106158. <https://doi.org/10.1016/j.apgeochem.2024.106158>.
- [60] Wang L, Huang X, Yu Y, Zhao L, Wang C, Feng Z, Cui D, Long Z. Towards cleaner production of rare earth elements from bastnaesite in China. *J Clean Prod.* 2017;165:231–42. <https://doi.org/10.1016/j.jclepro.2017.07.107>.
- [61] Fathollahzadeh H, Eksteen JJ, Kaksonen AH, Watkin ELJ. Role of microorganisms in bioleaching of rare earth elements from primary and secondary resources. *Appl Microbiol Biotechnol.* 2019;103(3):1043. <https://doi.org/10.1007/s00253-018-9526-z>.
- [62] Cen P, Bian X, Wu W, Li B. A sustainable green technology for separation and simultaneous recovery of rare earth elements and fluorine in bastnaesite concentrates. *Sep Purif*

- Technol. 2021;274:118380. <https://doi.org/10.1016/j.seppur.2021.118380>.
- [63] Kanazawa Y, Kamitani M. Rare earth minerals and resources in the world. *J Alloys Compd.* 2006;408:1339. <https://doi.org/10.1016/j.jallcom.2005.04.033>.
- [64] Eggert R, Wadia C, Anderson C, Bauer D, Fields F, Meinert L, Taylor P. Rare earths: market disruption, innovation, and global supply chains. *Annu Rev Environ Resour.* 2016. <https://doi.org/10.1146/annurev-environ-110615-085700>.
- [65] Harmer RE, Nex PAM. Rare earth deposits of Africa. *Episodes.* 2016;39(2):381–406. <https://doi.org/10.18814/epiugs/2016/v39i2/95784>.
- [66] McLeod CL, Shaulis BJ. Rare earth elements in planetary crusts: insights from chemically evolved igneous suites on earth and the moon. *Minerals.* 2018;8(10):455. <https://doi.org/10.3390/min8100455>.
- [67] Wang DH, Zhao Z, Yu Y, Dai JJ, Deng MC, Zhao T, Liu LJ. Exploration and research progress on ion-adsorption type REE deposit in South China. *China Geol.* 2018;1(3):415. <https://doi.org/10.31035/cg2018022>.
- [68] Spandler C, Slezak P, Nazari-Dehkordi T. Tectonic significance of Australian rare earth element deposits. *Earth-Sci Rev.* 2020;207:103219. <https://doi.org/10.1016/j.earscirev.2020.103219>.
- [69] Zhukova IA, Stepanov AS, Jiang SY, Murphy D, Mavrogenes J, Allen C, Chen W, Bottrill R. Complex REE systematics of carbonatites and weathering products from uniquely rich Mount Weld REE deposit, Western Australia. *Ore Geol Rev.* 2021;139:104539. <https://doi.org/10.1016/j.oregeorev.2021.104539>.
- [70] Liu SL, Fan HR, Liu X, Meng JY, Butcher AR, Yann L, Yang KF, Li XC. Global rare earth elements projects: new developments and supply chains. *Ore Geol Rev.* 2023;157:105428. <https://doi.org/10.1016/j.oregeorev.2023.105428>.
- [71] USGS National Minerals Information Center. Rare Earths Statistics and Information-Statistics and information on the worldwide supply of, demand for, and flow of the mineral commodity group rare earths-scandium, yttrium, and the lanthanides. 2023; <https://www.usgs.gov/centers/national-minerals-information-center/rare-earths-statistics-and-information>. Accessed 18 May 2025.
- [72] Statista. Total rare earth oxide demand worldwide excluding China from 2010 to 2025. 2023; <https://www.statista.com/statistics/450098/global-rare-earth-oxide-demand-outside-of-china-forecast/>. Accessed 18 May 2025.
- [73] Wang L, Wang P, Wang Q, Liu Y, Zhang B, Chen W. Global distribution and development potential of rare earth resources. *Sci Technol Rev.* 2022;40(8):27. <https://doi.org/10.3981/j.issn.1000-7857.2022.08.003>.
- [74] Chen Y, Zheng B. What happens after the rare earth crisis: a systematic literature review. *Sustainability.* 2019;11(5):1288. <https://doi.org/10.3390/su11051288>.
- [75] Statista. Demand for rare earth oxides worldwide in 2019 and 2025 by end use. 2023; <https://www.statista.com/statistics/449722/rare-earth-estimated-demand-globally-by-application/>. Accessed 18 May 2025.
- [76] Yu S, Duan H, Cheng J. An evaluation of the supply risk for China's strategic metallic mineral resources. *Resour Policy.* 2021;70:101891. <https://doi.org/10.1016/j.resourpol.2020.101891>.
- [77] Statista. Rare earth oxide prices worldwide in 2020 and a forecast for 2021 to 2030. 2023; <https://www.statista.com/statistics/449838/forecast-average-rare-earth-oxide-prices-globally/>. Accessed 18 May 2025.
- [78] Liu Y, Liang Y. Research progress and hot spot analysis of China's rare earth export based on CiteSpace software. *China Min Mag.* 2020;29(7):44. <https://doi.org/10.12075/j.issn.1004-4051.2020.07.026>.
- [79] Wu Y, Peng Z, Lai D, Zhao S, Wang L, Chen W, Wang P. Exploring international rare earth industry landscape changes and China's strategic responses. *Bull Chin Acad Sci.* 2023;38(2):255. <https://doi.org/10.16418/j.issn.1000-3045.20221026001>.
- [80] Wuebbeke J. Rare earth elements in China: policies and narratives of reinventing an industry. *Resour Policy.* 2013;38(3):384. <https://doi.org/10.1016/j.resourpol.2013.05.005>.
- [81] Zhao Z, Wang D, Bagas L, Chen Z. Geochemical and REE mineralogical characteristics of the Zhaibei Granite in Jiangxi Province, southern China, and a model for the genesis of ion-adsorption REE deposits. *Ore Geol Rev.* 2022;140:104579. <https://doi.org/10.1016/j.oregeorev.2021.104579>.
- [82] Huang XW, Long ZQ, Wang LS, Feng ZY. Technology development for rare earth cleaner hydrometallurgy in China. *Rare Met.* 2015;34(4):215. <https://doi.org/10.1007/s12598-015-0473-x>.
- [83] Borst AM, Smith MP, Finch AA, Estrade G, Villanova-de-Benavent C, Nason P, Marquis E, Horsburgh NJ, Goodenough KM, Xu C, Kynicky J, Geraki K. Adsorption of rare earth elements in regolith-hosted clay deposits. *Nat Commun.* 2020;11(1):4386. <https://doi.org/10.1038/s41467-020-17801-5>.
- [84] Zhou X, Wen D, Luo X. The status quo and development trend of the extraction technology of ion-absorbed rare earth in southern China. *Nonferrous Me Sci Eng.* 2012;3(6):81. <https://doi.org/10.13264/j.cnki.yjskx.2012.06.016>.
- [85] Kamprowski R. The raw materials policy of the European Union for rare earth elements. *Rocznik Integracji Europejskiej.* 2022;16:243. <https://doi.org/10.14746/rie.2022.16.14>.
- [86] Mancheri NA. World trade in rare earths, Chinese export restrictions, and implications. *Resour Policy.* 2015;46:262. <https://doi.org/10.1016/j.resourpol.2015.10.009>.
- [87] POLITICO. Sweden's rare earths discovery won't end EU's China reliance—yet. 2023. <https://www.politico.eu/article/swedens-rare-earths-discovery-wont-end-eus-china-reliance-yet/>. Accessed 18 May 2025.
- [88] MNRPRC (Ministry of Natural Resources of the People's Republic of China), 2016. Japan's deep sea rare earth development, the road is still far away! 2016. https://www.mnr.gov.cn/dt/kc/201611/t20161120_2321527.html. Accessed 18 May 2025.
- [89] Statista. Average Chinese domestic prices for selected rare earth oxides as of January 2023. 2023. <https://www.statista.com/statistics/280038/chinese-domestic-and-export-prices-for-rare-earth-oxides/>. Accessed 18 May 2025.
- [90] Shi Q, Zhao Y, Meng X, Shen L, Qiu G, Zhang X, Yu H, He X, He H, Zhao H. Column leaching of ion adsorption rare earth ore at low ammonium concentration. *J Mater Res Technol JMR&T.* 2022;19:2135. <https://doi.org/10.1016/j.jmrt.2022.05.199>.
- [91] Vahidi E, Navarro J, Zhao F. An initial life cycle assessment of rare earth oxides production from ion-adsorption clays. *Resour Conserv Recycl.* 2016;113:1. <https://doi.org/10.1016/j.resconrec.2016.05.006>.
- [92] Amato A, Becci A, Birloaga I, De Michelis I, Ferella F, Innocenzi V, Ippolito NM, Jimenez Gomez CP, Veglio F, Beolchini F. Sustainability analysis of innovative technologies for the rare earth elements recovery. *Renew Sustain Energy Rev.* 2019;106:41. <https://doi.org/10.1016/j.rser.2019.02.029>.
- [93] Moldoveanu GA, Papangelakis VG. An overview of rare-earth recovery by ion-exchange leaching from ion-adsorption clays of various origins. *Mineral Mag.* 2016;80(1):63. <https://doi.org/10.1180/minmag.2016.080.051>.

- [94] Feng Z, Huang X, Wang M, Zhang G. Progress and trend of green chemistry in extraction and separation of typical rare earth resources. *Chin J Rare Met.* 2017;41(5):604. <https://doi.org/10.13373/j.cnki.cjrm.XY17030028>.
- [95] Wu Z, Chen Y, Wang Y, Xu Y, Lin Z, Liang X, Cheng H. Review of rare earth element (REE) adsorption on and desorption from clay minerals: application to formation and mining of ion-adsorption REE deposits. *Ore Geol Rev.* 2023; 157:105446. <https://doi.org/10.1016/j.oregeorev.2023.105446>.
- [96] Hu Y, Wang L, Cao Z, Zhang W. Research progress on rare earth ore metallurgy and separation technology in China. *Conserv Util Miner Resour.* 2020;40:151. <https://doi.org/10.13779/j.cnki.issn1001-0076.2020.02.020>.
- [97] He Q, Chen J, Gan L, Gao M, Zan M, Xiao Y. Insight into leaching of rare earth and aluminum from ion adsorption type rare earth ore: adsorption and desorption. *J Rare Earths.* 2023; 41(9):1398. <https://doi.org/10.1016/j.jre.2022.08.009>.
- [98] Meng X, Zhao H, Zhao Y, Shen L, Gu G, Qiu G. Heap leaching of ion adsorption rare earth ores and REEs recovery from leachate with lixiviant regeneration. *Sci Total Environ.* 2023; 898:165417. <https://doi.org/10.1016/j.scitotenv.2023.165417>.
- [99] Xiao YF, Chen YY, Feng ZY, Huang XW, Huang L, Long ZQ, Cui DL. Leaching characteristics of ion-adsorption type rare earths ore with magnesium sulfate. *Trans Nonferrous Met Soc China.* 2015;25(11):3784. [https://doi.org/10.1016/s1003-6326\(15\)64022-5](https://doi.org/10.1016/s1003-6326(15)64022-5).
- [100] Xiao Y, Huang X, Feng Z, Liu X, Huang L, Long Z. Environmental impact assessment and prospect for ion-adsorption type rare earths ore leaching by magnesium salt. *J Chin Soc Rare Earths.* 2015;33(1):1. <https://doi.org/10.11785/S1000-4343.20150101>.
- [101] Xiao Y, Feng Z, Huang X, Huang L, Chen Y, Liu X, Wang L, Long Z. Recovery of rare earth from the ion-adsorption type rare earths ore: II. Compound leaching. *Hydrometallurgy.* 2016;163:83. <https://doi.org/10.1016/j.hydromet.2016.03.016>.
- [102] Xie D, Fu C, Tang Z, Liu X. Current status of rare earth resources in China and progress of extracting technology. *Conserv Utiliz Min Resour.* 2021;41(1):152. <https://doi.org/10.13779/j.cnki.issn1001-0076.2021.01.022>.
- [103] Lai F, Gao G, Huang L, Xiao Y, Yang R, Li K. Compound leaching of rare earth from the ion-adsorption type rare earth ore with magnesium sulfate and ascorbic acid. *Hydrometallurgy.* 2018;179:25. <https://doi.org/10.1016/j.hydromet.2018.05.027>.
- [104] Zhao M, Teng Z, Ma X, Jiang X, Zhang H, Yang Y, Li T. Insight into leaching rare earth from ion-adsorption type rare earth ores with citric acid: performance, kinetic analysis and differentiation leaching. *J Rare Earths.* 2025;43(3):591. <https://doi.org/10.1016/j.jre.2024.02.005>.
- [105] Lai F, Huang L, Gao G, Yang R, Xiao Y. Recovery of rare earths from ion-absorbed rare earths ore with MgSO₄-ascorbic acid compound leaching agent. *J Rare Earths.* 2018;36(5):521. <https://doi.org/10.1016/j.jre.2017.12.003>.
- [106] Hu Z, Zhang Z, Ra C, Chen Z, Chi X, Liu D. Enhancement of the leaching process of weathered crust elution-deposited rare earth ore with composite magnesium salt. *Metal Min.* 2020; 2020(03):95. <https://doi.org/10.13779/j.cnki.issn1001-0076.2020.02.020>.
- [107] Zhao M, Zhang H, Han H, Jiang X, Yang Y, Li T. Differential leaching mechanisms and ecological impact of organic acids on ion-adsorption type rare earth ores. *Sep Purif Technol.* 2025; 362:131701. <https://doi.org/10.1016/j.seppur.2025.131701>.
- [108] Li S, Jansone-Popova S, Jiang DE. Insights into coordination and ligand trends of lanthanide complexes from the Cambridge structural database. *Sci Rep.* 2024;14(1):11301. <https://doi.org/10.1038/s41598-024-62074-3>.
- [109] Pathak A, Al-Sheeha H, Navvamani R, Kothari R, Marafi M, Rana MS. Recycling of platinum group metals from exhausted petroleum and automobile catalysts using bioleaching approach: a critical review on potential, challenges, and outlook. *Rev Environ Sci Bio-Technol.* 2022;21(4):1035. <https://doi.org/10.1007/s11157-022-09636-x>.
- [110] Wang W, Liu L, Liu H, Zhang B, Cao Y, Wang H. Progress and trend of rare earth resources extraction technology. *Conserv Util Min Resour.* 2020;40(5):32. <https://doi.org/10.13779/j.cnki.issn1001-0076.2020.05.005>.
- [111] Yang X, Zhang Z, Kuang S, Wei H, Li Y, Wu G, Geng A, Li Y, Liao W. Removal of thorium and uranium from leach solutions of ion-adsorption rare earth ores by solvent extraction with Cextrant 230. *Hydrometallurgy.* 2020;194:105343. <https://doi.org/10.1016/j.hydromet.2020.105343>.
- [112] Lee JCK, Wen Z. Rare earths from mines to metals comparing environmental impacts from China's main production pathways. *J Ind Ecol.* 2017;21(5):1277. <https://doi.org/10.1111/jiec.12491>.
- [113] Shen L, Zhou H, Shi Q, Meng X, Zhao Y, Qiu G, Zhang X, Yu H, He X, He H, Zhao H. Comparative chemical and non-contact bioleaching of ion-adsorption type rare earth ore using ammonium sulfate and metabolites of *Aspergillus niger* and *Yarrowia lipolytica* to rationalise the role of organic acids for sustainable processing. *Hydrometallurgy.* 2023;216: 106019. <https://doi.org/10.1016/j.hydromet.2023.106019>.
- [114] Stockdale A, Banwart SA. Recovery of technologically critical lanthanides from ion adsorption soils. *Miner Eng.* 2021;168: 106921. <https://doi.org/10.1016/j.mineng.2021.106921>.
- [115] Cen P, Bian X, Liu Z, Gu M, Wu W, Li B. Extraction of rare earths from bastnaesite concentrates: a critical review and perspective for the future. *Miner Eng.* 2021;171:107081. <https://doi.org/10.1016/j.mineng.2021.107081>.
- [116] Zhang Z, Jiang L, Guo W, Yang J, Liu D, Chi Ra. Flotation behavior and adsorption mechanism of 2-hydroxy-3-naphthyl hydroxamic acid on bastnaesite surface. *J Rare Earths.* 2025; 43(5):1084. <https://doi.org/10.1016/j.jre.2024.04.007>.
- [117] Liu Q, Gao P, Liu J, Guo C, Wen G. Flotation performance and adsorption mechanism of monazite after the transformation of hydrogen-based mineral phase by octyl hydroxamic acid. *Powder Technol.* 2025;451:120462. <https://doi.org/10.1016/j.powtec.2024.120462>.
- [118] Abaka-Wood GB, Xu S, Ayedzi LD, Addai-Mensah J, Skinner W. Flotation recovery of monazite from kaolinite using sodium oleate collector: understanding mineral-collector interaction. *Miner Eng.* 2024;209:108605. <https://doi.org/10.1016/j.mineng.2024.108605>.
- [119] Zhang Q, Sun Y, Gao P, Han Y. Hydrogen-based mineral phase transformation of bastnaesite: detailed assessment of physico-chemical properties and flotation behavior. *Chem Eng J.* 2024; 500:156992. <https://doi.org/10.1016/j.cej.2024.156992>.
- [120] Bazin C, Boulanger J-F. Investigation of the flotation of an ore containing bastnaesite and monazite: kinetic study and process flowsheet simulation. *Minerals.* 2024;14(9):906. <https://doi.org/10.3390/min14090906>.
- [121] Liu C, Xu L, Deng J, Han Z, Tian J, Wang D, Xue K, Wang Z, Shu K. Multi-polar group cationic collector HTTPD: mechanistic insights into selective flotation of bastnaesite. *Chem Eng J.* 2024;498:155806. <https://doi.org/10.1016/j.cej.2024.155806>.
- [122] Lin P, Yang X, Werner JM, Honaker RQ. Application of Eh-pH diagrams on acid leaching systems for the recovery of REEs from bastnaesite, monazite and xenotime. *Metals.* 2021; 11(5):734. <https://doi.org/10.3390/met11050734>.
- [123] Huang X, Li H, Long Z, Liu Y, Zhao N, Zhang G. A smelting method for iron-rich monazite rare earth ore. *China Patent;* 00710146235.1, 2009.

- [124] Zhang H, Xie D, Zhang H, Jiang M, Huang H, Wan Y, Liao Y, Zhao S. Reaction mechanism of fluoride conversion into BF₃ during sulphuric acid leaching of roasted bastnaesite. *J Rare Earths*. 2021;39(2):186. <https://doi.org/10.1016/j.jre.2020.03.005>.
- [125] Colligan G, O'Kelley B, Anderson C. Novel methods for bastnaesite concentrate leaching. *Min Metall Explor*. 2022; 39(1):31. <https://doi.org/10.1007/s42461-021-00525-5>.
- [126] Cheng S, Li W, Han Y, Sun Y, Gao P, Zhang X. Recent process developments in beneficiation and metallurgy of rare earths: a review. *J Rare Earths*. 2024;42(4):629. <https://doi.org/10.1016/j.jre.2023.03.017>.
- [127] Yörükoglu A, Obut A, Girgin I. Effect of thiourea on sulphuric acid leaching of bastnaesite. *Hydrometallurgy*. 2003;68(1–3): 195. [https://doi.org/10.1016/s0304-386x\(02\)00199-8](https://doi.org/10.1016/s0304-386x(02)00199-8).
- [128] Shahbaz A. A systematic review on leaching of rare earth metals from primary and secondary sources. *Miner Eng*. 2022; 184:107632. <https://doi.org/10.1016/j.mineng.2022.107632>.
- [129] Liu M, Guo L, He H. Research and development of technology for rare earth metal leaching and recovery. *Ind Min Process*. 2023;52(9):19. <https://doi.org/10.16283/j.cnki.hgkwyjg.2023.09.004>.
- [130] Wang L, Wang C, Li L, Yang YM. Readsorption of rare earth elements during leaching process of ion-adsorption-type rare earth ore. *Rare Met*. 2023;42(6):2113. <https://doi.org/10.1007/s12598-018-1162-3>.
- [131] Huang Y, Zhang T-a, Dou Z, Han G, Cao Y, Hou C. Decomposition mechanism of a mixed rare earth concentrate with sodium hydroxide in the microwave heating process. *Miner Eng*. 2019;132:220. <https://doi.org/10.1016/j.mineng.2018.12.021>.
- [132] Yang Y, Liu Y, Huang C-F, de Silva J, Zhao F-J. Aluminium alleviates fluoride toxicity in tea (*Camellia sinensis*). *Plant Soil*. 2016;402(1–2):179. <https://doi.org/10.1007/s11104-015-2787-8>.
- [133] Choubisa SL. A brief review of fluoride-induced bone disease skeletal fluorosis in humans and its prevention. *J Pharm Pharmacol Res*. 2024;7(8):200. <https://doi.org/10.31579/2688-7517/200>.
- [134] Taher MK, Momoli F, Go J, Hagiwara S, Ramoju S, Hu X, Jensen N, Terrell R, Hemmerich A, Krewski D. Systematic review of epidemiological and toxicological evidence on health effects of fluoride in drinking water. *Crit Rev Toxicol*. 2024; 54(1):2. <https://doi.org/10.1080/10408444.2023.2295338>.
- [135] Zhang XW, Li M, Liu ZG, Hu YH, Wang MT. Kinetics of rare earth extraction from Baotou bastnaesite in hydrochloric acid and aluminum chloride. *JOM*. 2017;69(10):1894. <https://doi.org/10.1007/s11837-017-2379-6>.
- [136] He J, Li Y, Xue X, Ru H, Huang X, Yang H. Leaching of fluorine and rare earths from bastnaesite calcined with aluminum hydroxide and the recovery of fluorine as cryolite. *RSC Adv*. 2017;7(23):14053. <https://doi.org/10.1039/c6ra28106k>.
- [137] Liu W, Feng X, Noble A, Yoon RH. Ammonium sulfate leaching of NaOH-treated monazite. *Miner Eng*. 2022;188: 107817. <https://doi.org/10.1016/j.mineng.2022.107817>.
- [138] Demol J, Ho E, Soldenhoff K, Karatchevtseva I, Senanayake G. Beneficial effect of iron oxide/hydroxide minerals on sulfuric acid baking and leaching of monazite. *Hydrometallurgy*. 2022; 211:105864. <https://doi.org/10.1016/j.hydromet.2022.105864>.
- [139] Watts H, Fisher T. Leaching the unleachable mineral: rare earth dissolution from monazite ore in condensed phosphoric acid. *Minerals*. 2021;11(9):931. <https://doi.org/10.3390/min11090931>.
- [140] Demol J, Ho E, Soldenhoff K, Senanayake G. The impact of apatite on sulfuric acid baking of a monazite concentrate and the benefit of goethite addition on rare earth leaching. *Hydrometallurgy*. 2024;226:106296. <https://doi.org/10.1016/j.hydromet.2024.106296>.
- [141] Chen S, Feng Z, Wang M, Zhao L, Yu Z, Xia C, Huang X. Leaching kinetic study of sulfuric acid roasted mixed-type rare earth concentrate for reducing the solid-waste production and chemical consumption. *J Clean Prod*. 2020;260:120989. <https://doi.org/10.1016/j.jclepro.2020.120989>.
- [142] Qiao S, Bian X, Cui J, Cen P, Wu W. Clean metallurgy and technical progress of light rare earth minerals. *Can Metall Q*. 2024;63(2):538. <https://doi.org/10.1080/00084433.2023.2193522>.
- [143] Hou S, Huang W, Liu Y, Zhang B, Liu C. Effect of phosphoric acid on leaching of monazite during low-temperature sulfuric acid cyclic leaching process. *J Rare Earths*. 2025;43(4):784. <https://doi.org/10.1016/j.jre.2024.04.003>.
- [144] Tunsu C, Menard Y, Eriksen DO, Ekberg C, Petranikova M. Recovery of critical materials from mine tailings: a comparative study of the solvent extraction of rare earths using acidic, solvating and mixed extractant systems. *J Clean Prod*. 2019; 218:425. <https://doi.org/10.1016/j.jclepro.2019.01.312>.
- [145] Ilyas S, Srivastava RR, Kim H. Solvent extraction for high separation strategy of light and heavy rare earth elements from a sulfate-leached solution of low-grade monazite. *Sep Purif Technol*. 2024;341:126896. <https://doi.org/10.1016/j.seppur.2024.126896>.
- [146] Han J, Li Y, Li S, Liao W. Efficient enrichment and selective recovery of rare earths from sulfate leachate of ion-adsorption type rare earth ore by extraction with HPOAc. *J Rare Earths*. 2024;42(11):2156. <https://doi.org/10.1016/j.jre.2023.10.018>.
- [147] Wang C, Harbottle D, Liu Q, Xu Z. Current state of fine mineral tailings treatment: a critical review on theory and practice. *Miner Eng*. 2014;58:113. <https://doi.org/10.1016/j.mineng.2014.01.018>.
- [148] Tan Q, Li J, Zeng X. Rare earth elements recovery from waste fluorescent lamps: a review. *Crit Rev Environ Sci Technol*. 2015;45(7):749. <https://doi.org/10.1080/10643389.2014.900240>.
- [149] Vuppalladadiyam SSV, Thomas BS, Kundu C, Vuppalladadiyam AK, Duan H, Bhattacharya S. Can e-waste recycling provide a solution to the scarcity of rare earth metals? An overview of e-waste recycling methods. *Sci Total Environ*. 2024;924:171453.
- [150] Favot M, Massarutto A. Rare-earth elements in the circular economy: the case of yttrium. *J Environ Manag*. 2019;240:504. <https://doi.org/10.1016/j.jenvman.2019.04.002>.
- [151] Sethurajan M, van Hullebusch ED, Fontana D, Akcil A, Deveci H, Batinic B, Leal JP, Gasche TA, Kucuker MA, Kuchta K, Neto IFF, Soares HMVM, Chmielarz A. Recent advances on hydrometallurgical recovery of critical and precious elements from end of life electronic wastes—a review. *Crit Rev Environ Sci Technol*. 2019;49(3):212. <https://doi.org/10.1080/10643389.2018.1540760>.
- [152] Dutta D, Rautela R, Gujjala LKS, Kundu D, Sharma P, Tembhare M, Kumar S. A review on recovery processes of metals from E-waste: a green perspective. *Sci Total Environ*. 2023; 859:160391. <https://doi.org/10.1016/j.scitotenv.2022.160391>.
- [153] Danouche M, Bounaga A, Oulkhair A, Boulif R, Zeroual Y, Benhida R, Lyamlouli K. Advances in bio/chemical approaches for sustainable recycling and recovery of rare earth elements from secondary resources. *Sci Total Environ*. 2024;912: 168811. <https://doi.org/10.1016/j.scitotenv.2023.168811>.
- [154] Gkika DA, Chalaris M, Kyzas GZ. Review of methods for obtaining rare earth elements from recycling and their impact on the environment and human health. *Processes*. 2024;12(6): 1235. <https://doi.org/10.3390/pr12061235>.

- [155] O'Connell-Danes JG, Ilik BO, Hull EE, Ngwenya BT, Morrison CA, Love JB. A simple supramolecular approach to recycling rare earth elements. *ACS Sustain Chem Eng*. 2024; 12(25):9301. <https://doi.org/10.1021/acssuschemeng.4c03063>.
- [156] Zhang B, Wu J, Mou G, Xiao M, Chu S, Yang Z. Evaluation on rare earth elements and microbial communities in abandoned rare earth tailings. *J Geochem Explor*. 2025;272:107715. <https://doi.org/10.1016/j.gexplo.2025.107715>.
- [157] Yang Y, Zhang M, Feng L, Huang B, Zhai R, Sun X. Sustainable reutilization of ion-adsorbed rare earth tailings: preparation of low-cost functionalized pigments. *Ceram Int*. 2024;50(7):11575. <https://doi.org/10.1016/j.ceramint.2024.01.058>.
- [158] Medina-Díaz HL, López-Bellido FJ, Alonso-Azcárate J, Fernández-Morales FJ, Rodríguez L. Can rare earth elements be recovered from abandoned mine tailings by means of electrokinetic-assisted phytoextraction? *Environ Sci Pollut Res*. 2024; 31(18):26747. <https://doi.org/10.1007/s11356-024-32759-3>.
- [159] Ge J, Lei Y. Resource tax on rare earths in China: policy evolution and market responses. *Resour Policy*. 2018;59:291. <https://doi.org/10.1016/j.resourpol.2018.07.016>.
- [160] Packey DJ, Kingsnorth D. The impact of unregulated ionic clay rare earth mining in China. *Resour Policy*. 2016;48:112. <https://doi.org/10.1016/j.resourpol.2016.03.003>.
- [161] MNRPRC (Ministry of Natural Resources of the People's Republic of China), 2014. Consolidation in the rare earth industry will be accelerated. 2014. https://www.mnr.gov.cn/dt/pl/201402/t20140227_2347109.html.
- [162] MNRPRC (Ministry of Natural Resources of the People's Republic of China). Progress has been made in the management, governance and rectification of the rare earth industry. 2015. https://www.mnr.gov.cn/dt/kc/201508/t20150812_2320608.html. Accessed 18 May 2025.
- [163] MNRPRC (Ministry of Natural Resources of the People's Republic of China). Satellite imagery law enforcement of natural resources will be fully launched to help ecological and environmental protection Consolidation in the rare earth industry may accelerate. 2023. https://www.mnr.gov.cn/dt/ywbb/202303/t20230320_2778865.html. Accessed 18 May 2025.
- [164] MNRPRC (Ministry of Natural Resources of the People's Republic of China), 2023. Farmland, ecology and rare earth protection will be the focus of law enforcement in 2023. 2023. https://www.mnr.gov.cn/dt/ywbb/202302/t20230224_2776721.html. Accessed 18 May 2025.
- [165] Chen Z, Hu Z, Li K. The spillover effect of trade policy along the value chain: evidence from China's rare earth-related sectors. *World Econ*. 2021;44(12):3550. <https://doi.org/10.1111/twec.13172>.
- [166] Statista. Rare earths industry worldwide. 2023. <https://www.statista.com/study/10040/rare-earths-statista-dossier/>. Accessed 18 May 2025.
- [167] Liang T, Li K, Wang L. State of rare earth elements in different environmental components in mining areas of China. *Environ Monit Assess*. 2014;186(3):1499. <https://doi.org/10.1007/s10661-013-3469-8>.
- [168] Brown M, Eggert R. Simulating producer responses to selected Chinese rare earth policies. *Resour Policy*. 2018;55:31. <https://doi.org/10.1016/j.resourpol.2017.10.013>.
- [169] Ho P, Zhao H. Mining conflict and rent-seeking in China: a mixed method analysis of cases of illegality. *Extr Ind Soc*. 2022;9:101031. <https://doi.org/10.1016/j.exis.2021.101031>.
- [170] Diao H, Yang H, Tan T, Ren G, You M, Wu L, Yang M, Bai Y, Xia S, Song S, Quintana M, Liu L, Xue Q. Navigating the rare earth elements landscape: challenges, innovations, and sustainability. *Miner Eng*. 2024;216:108889. <https://doi.org/10.1016/j.mineng.2024.108889>.
- [171] Yin X, Ang LP, Chang ZY. Rare earth element mining threatens Malaysia's biodiversity. *Science*. 2024;384(6701):1182. <https://doi.org/10.1126/science.adp2846>.
- [172] Xiao W, Zhou L, Yang P, Yan N, Wei C. An international comparative study of rare earth research from the perspective of bibliometrics. *Heliyon*. 2023;9(5):e16075. <https://doi.org/10.1016/j.heliyon.2023.e16075>.
- [173] Zheng B, Zhang YW, Geng Y, Wei W, Tan X, Xiao S, Gao Z. Measuring the anthropogenic cycles of light rare earths in China: implications for the imbalance problem. *Sci Total Environ*. 2023;879:163215. <https://doi.org/10.1016/j.scitotenv.2023.163215>.
- [174] Leng Z, Sun H, Cheng J, Wang H, Yao Z. China's rare earth industry technological innovation structure and driving factors: a social network analysis based on patents. *Resour Policy*. 2021;73:102233. <https://doi.org/10.1016/j.resourpol.2021.102233>.
- [175] Shuai J, Zhao Y, Shuai C, Wang J, Yi T, Cheng J. Assessing the international co-opetition dynamics of rare earth resources between China, USA, Japan and the EU: an ecological niche approach. *Resour Policy*. 2023;82:103446. <https://doi.org/10.1016/j.resourpol.2023.103446>.
- [176] Zhang HM, Feng TT, Yang YS. Influencing factors and critical path of inter-sector embodied heavy rare earth consumption in China. *Resour Policy*. 2022;75:102492. <https://doi.org/10.1016/j.resourpol.2021.102492>.
- [177] Guo Q, You W. A comprehensive evaluation of the international competitiveness of strategic minerals in China, Australia, Russia and India: the case of rare earths. *Resour Policy*. 2023; 85:103821. <https://doi.org/10.1016/j.resourpol.2023.103821>.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.